

MARTI101

AI and Digital Business Transformation

EU Business School - Masters Level 7 - Term I - 3 ECTS

Glossary Reference Library

Master Index - 6 Units - 48 Definition Sheets

6 **Course Units**
one glossary per unit, colour-coded

48 **Professional Definitions**
eight per unit

48 **Reference Sheets**
print-ready A4, nine sections each

175+ **Harvard-Referenced Sources**
peer-reviewed, statutory and practitioner

Design Philosophy

Every sheet follows one nine-section layout: the professional definition, who defined it originally, why the definition was developed, a custom figure, how, when and by whom the term is used, what it improved on relative to its predecessor tool, the related definitions elsewhere in MARTI101, and Harvard-referenced sources. The constant structure lets the library be read as a reference work rather than a sequence: each sheet is a thumbnail in the sense of Meier (2025), navigable by unit, by keyword, or by following the related-definition pills from one sheet to the next.

Master Index

Six unit glossaries - 48 definition sheets

UNIT 1 - Digital Transformation & Artificial Intelligence

8 sheets

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| 1 Digital Transformation | 5 Business Model Innovation |
| 2 Artificial Intelligence | 6 Disruptive Innovation |
| 3 Digital Maturity | 7 Technology Adoption Lifecycle |
| 4 Digitisation and Digitalisation | 8 Value Migration |

UNIT 2 - Business Leadership in the AI Era

8 sheets

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| 1 AI Literacy | 5 Change Leadership |
| 2 Psychological Safety | 6 Human-in-the-Loop |
| 3 Upper Echelons Theory | 7 Algorithm Aversion |
| 4 Organisational Culture | 8 Decision Rights |

UNIT 3 - Business Strategies in the AI Era

8 sheets

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| 1 AI Strategy | 5 Build-Buy-Partner Decision |
| 2 Data Strategy | 6 Scalable AI Business Model |
| 3 Competitive Advantage | 7 Profit Pool |
| 4 AI Operating Model | 8 AI Portfolio Management |

Master Index

Six unit glossaries - 48 definition sheets

UNIT 4 - Leading Organisational AI Adoption

8 sheets

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| 1 AI Adoption | 5 MLOps |
| 2 AI Roadmap | 6 Reskilling and Upskilling |
| 3 Proof of Concept and Pilot Purgatory | 7 Technology Acceptance Model |
| 4 Change Management | 8 Benefits Realisation |

UNIT 5 - AI's Ripple Effects: Business, Economy & Environment

8 sheets

- | | |
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| 1 Productivity Paradox | 5 Externality |
| 2 General Purpose Technology | 6 Green AI |
| 3 Skill-Biased Technological Change | 7 Life Cycle Assessment |
| 4 Labour Market Displacement | 8 Digital Divide |

UNIT 6 - Human-Centred Implications of AI: Ethics, Society & Regulation

8 sheets

- | | |
|--------------------|-----------------------------|
| 1 AI Ethics | 5 AI Governance |
| 2 EU AI Act | 6 Accountability Gap |
| 3 Algorithmic Bias | 7 Data Protection by Design |
| 4 Explainability | 8 Trustworthy AI |

How to Use This Library

For students

Each sheet answers nine fixed questions about one term. Read the professional definition and the figure first; together they take under two minutes and are sufficient for recognition. Read the remaining sections when you need to use the term rather than recognise it. In assessed work cite the primary source listed at the foot of the sheet, never this library.

For practising executives

The library is a decision-support reference. When facing an AI investment, adoption or governance decision, locate the relevant term by unit or keyword and read the how, when and by-whom columns; these state the conditions under which the tool applies and the roles that own it. The innovation-relative-to-predecessor section explains what problem the term was created to solve, which is usually the fastest test of whether it fits your situation.

For MARTI101 assessments

Unit briefs expect the explicit application of named terms to the case at hand. Students should cite the specific sheet, for example 'Applying value migration (Glossary 1.8) to the Nordwerk case', and then the primary source behind it. Definitional precision is assessed: the digitisation/digitalisation and upskilling/reskilling distinctions in particular are marked.

Reading order

Linear reading follows the course sequence and is recommended on first encounter. Thereafter use the related-definitions pills at the foot of each sheet, which link forward and backward across units - for example from Business Model Innovation in Unit 1 to Scalable AI Business Model in Unit 3 and Profit Pool in the same unit.

Design conventions

Colour signals the unit: indigo for Unit 1, violet for Unit 2, blue for Unit 3, amber for Unit 4, green for Unit 5, red for Unit 6. Each sheet's accent bar and title carry its unit's colour. Harvard author-date referencing is used throughout, with full citations at the foot of every sheet. A red verification block marks any sheet whose practitioner figures vary materially by source; cross-check those against the primary report before assessment use.

Digital Transformation

The use of technology to radically improve performance or reach of an enterprise

GLOSSARY 01 - SHEET 1 / 8

PROFESSIONAL DEFINITION

Digital transformation is the use of digital technology to radically improve the performance or reach of an enterprise, achieved by combining investment in technology (digital intensity) with the leadership capability required to drive change through the organisation (transformation-management intensity). It is distinguished from technology deployment by the requirement that operating model, customer experience and business processes are redesigned, not merely automated (Westerman, Bonnet and McAfee, 2014; Vial, 2019).

WHO DEFINED IT ORIGINALLY

The working definition used in this course is Westerman, Bonnet and McAfee's (2014) formulation, developed at the MIT Center for Digital Business with Capgemini Consulting from a study of 391 large firms. Vial (2019) later consolidated 282 works into an integrative process definition that frames transformation as a process in which digital technologies trigger structural change and generate value. Earlier antecedents include Venkatraman's (1994) IT-enabled business transformation levels.

WHY THIS DEFINITION WAS DEVELOPED

Firms were spending heavily on technology and reporting no financial improvement. The MIT-Capgemini team needed a construct that separated firms buying technology from firms changing how they work, in order to explain why investment and performance were uncorrelated. Defining transformation as a two-dimensional capability - technology plus governance - made the difference measurable and gave boards a diagnostic rather than a slogan.

HOW IT IS USED

Score the firm on digital intensity and on transformation-management intensity, place it in one of four quadrants, and use the gap between the two scores to decide whether the next euro should be spent on technology or on governance.

WHEN IT IS USED

At the opening of a transformation programme, at annual strategy review, when a portfolio of pilots has failed to reach production, and whenever a new leadership team inherits a technology agenda it did not set.

BY WHOM IT IS USED

Chief executives and boards setting the transformation mandate, chief digital and information officers owning delivery, and business unit heads accountable for the operating model that must actually change.

FIGURE 1 - THE TWO INTENSITIES OF TRANSFORMATION

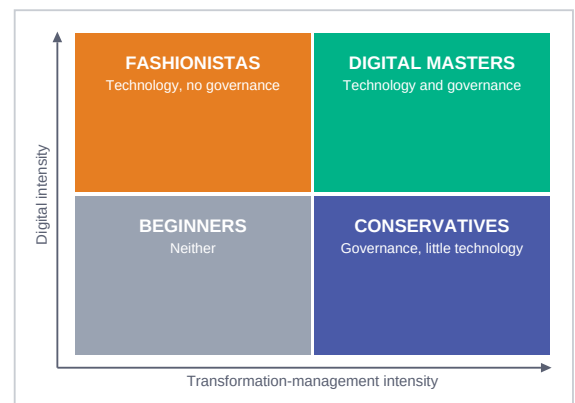


Figure 1. Digital transformation as the joint product of two intensities. Only the upper-right quadrant is associated with superior revenue generation and profitability; the Fashionista quadrant is the characteristic failure mode of technology-led programmes (after Westerman, Bonnet and McAfee, 2014).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor construct was IT-enabled business transformation (Venkatraman, 1994), which ranked change on a single ladder from localised exploitation to business scope redefinition. Westerman and colleagues replaced the ladder with two orthogonal axes, which allowed the empirically important failure case - heavy technology investment with weak governance, the Fashionista - to be named and observed. Vial's (2019) process definition then added the feedback loop: transformation changes the environment that triggered it.

RELATED DEFINITIONS IN THIS COURSE

Digital Maturity - Unit 1

Digitisation vs Digitalisation - Unit 1

Business Model Innovation - Unit 1

AI Adoption - Unit 4

AI Strategy - Unit 3

SOURCES (HARVARD REFERENCING STYLE)

- Venkatraman, N. (1994) 'IT-enabled business transformation: from automation to business scope redefinition', Sloan Management Review, 35(2), pp. 73-87.
- Vial, G. (2019) 'Understanding digital transformation: a review and a research agenda', Journal of Strategic Information Systems, 28(2), pp. 118-144.
- Westerman, G., Bonnet, D. and McAfee, A. (2014) Leading Digital: Turning Technology into Business Transformation. Boston, MA: Harvard Business Review Press.
- Westerman, G., Calmejsane, C., Bonnet, D., Ferraris, P. and McAfee, A. (2011) Digital Transformation: A Roadmap for Billion-Dollar Organizations. Cambridge, MA: MIT Center for Digital Business and Capgemini Consulting.

Artificial Intelligence

GLOSSARY 01 - SHEET 2 / 8

Machine-based systems that infer, from input, how to generate outputs that influence environments

PROFESSIONAL DEFINITION

An artificial intelligence system is a machine-based system that, for explicit or implicit objectives, infers from the input it receives how to generate outputs such as predictions, content, recommendations or decisions that can influence physical or virtual environments; different AI systems vary in their levels of autonomy and adaptiveness after deployment (OECD, 2024; Regulation (EU) 2024/1689, Article 3(1)).

WHO DEFINED IT ORIGINALLY

The definition in force for European business was drafted by the OECD's expert group for the 2019 Recommendation of the Council on Artificial Intelligence, revised in November 2023, and adopted with minor changes as the legal definition in Article 3(1) of the EU AI Act. The intellectual lineage runs to McCarthy, Minsky, Rochester and Shannon's (1955) Dartmouth proposal, which named the field.

WHY THIS DEFINITION WAS DEVELOPED

Regulators and standard-setters needed a definition that was technology-neutral and durable: narrow enough to exclude ordinary deterministic software, broad enough to survive the next architecture. Anchoring on inference - the system derives its outputs rather than having them enumerated by a programmer - achieved this, and the OECD-EU alignment gave firms one definition to satisfy across jurisdictions.

HOW IT IS USED

Apply the inference test to each candidate system to establish whether it falls inside the regulatory perimeter; systems that merely execute rules written by a human are outside it and need no conformity work.

WHEN IT IS USED

At procurement, at the start of any AI project intake, when compiling the AI system inventory required for governance, and whenever a vendor markets a product as artificial intelligence.

BY WHOM IT IS USED

Legal and compliance functions determining regulatory scope, procurement teams assessing vendor claims, and technology leaders maintaining the enterprise AI register.

FIGURE 1 - FROM INPUT TO ENVIRONMENTAL EFFECT

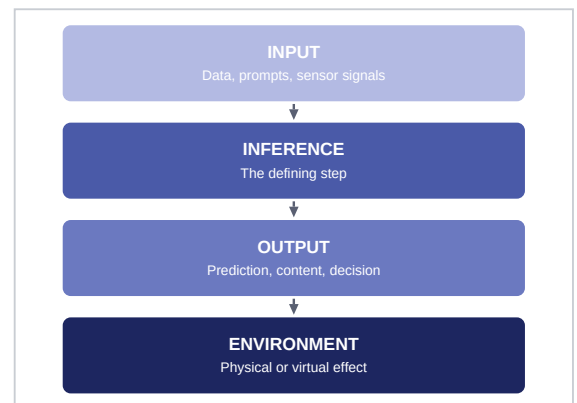


Figure 1. The inference test. A system is an AI system when the middle step is present: outputs are derived from input rather than enumerated in advance. Autonomy and adaptiveness vary along this chain and determine the obligations that attach to it (after OECD, 2024).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor definitions were capability lists - systems that perform tasks normally requiring human intelligence - which shifted every time a task was automated, a problem long known as the AI effect. The OECD formulation replaces the moving human benchmark with a structural test based on inference and autonomy, so classification does not change as public expectations change. Its adoption into binding EU law gave the definition legal consequence rather than descriptive convenience.

RELATED DEFINITIONS IN THIS COURSE

[Digital Transformation - Unit 1](#)[AI Literacy - Unit 2](#)[General Purpose Technology - Unit 5](#)[EU AI Act - Unit 6](#)[Human-in-the-Loop - Unit 2](#)

SOURCES (HARVARD REFERENCING STYLE)

European Parliament and Council (2024) Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence. Official Journal of the European Union, L, 12 July 2024.

McCarthy, J., Minsky, M.L., Rochester, N. and Shannon, C.E. (1955) A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence. Hanover, NH: Dartmouth College.

OECD (2024) Explanatory Memorandum on the Updated OECD Definition of an AI System. OECD Artificial Intelligence Papers No. 8. Paris: OECD Publishing.

Russell, S. and Norvig, P. (2021) Artificial Intelligence: A Modern Approach. 4th edn. Harlow: Pearson.

Digital Maturity

GLOSSARY 01 - SHEET 3 / 8

The degree to which a firm has aligned technology, capability and governance to a digital operating model

PROFESSIONAL DEFINITION

Digital maturity is the extent to which an organisation has developed and aligned the technological assets, human capabilities, processes and governance mechanisms required to compete through digital means. It is expressed as a position on an ordered scale of stages rather than as a binary state, and it is measured across multiple dimensions simultaneously so that unevenness between them becomes visible (Kane et al., 2017; Chanias and Hess, 2016).

WHO DEFINED IT ORIGINALLY

The stage-model tradition originates with Nolan's (1973) stages of data-processing growth and was adapted for the digital era by Kane, Palmer, Phillips, Kiron and Buckley (2017) through the MIT Sloan Management Review and Deloitte global survey, and independently by Chanias and Hess (2016) at Ludwig-Maximilians-Universitat Munchen in a comparative review of maturity instruments.

WHY THIS DEFINITION WAS DEVELOPED

Boards asked how far along they were and received answers expressed in project counts. A maturity construct was needed to convert activity into position, to make comparison across business units and against competitors possible, and to expose the common pathology in which technology maturity runs ahead of talent and governance maturity.

HOW IT IS USED

Assess each dimension separately against stage descriptors, plot the profile, and direct investment at the lagging dimension rather than at the strongest one, because maturity advances only at the rate of its weakest component.

WHEN IT IS USED

As a baseline before a transformation programme, annually thereafter to track movement, and at post-merger integration to compare the maturity of the two organisations being combined.

BY WHOM IT IS USED

Transformation programme offices running the assessment, executive committees allocating investment between dimensions, and business unit leaders benchmarking against the group profile.

FIGURE 1 - THE MATURITY STAIRCASE

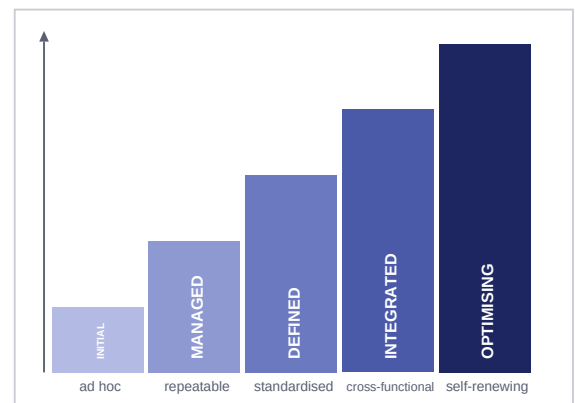


Figure 1. Maturity as an ordered progression. Each stage presupposes the one below it, and a firm advances only at the rate of its least mature dimension - the reason technology-first programmes stall at the third step (after Kane et al., 2017).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Nolan's original model assumed a single sequence through which every organisation passed at its own speed, and was criticised for weak empirical support. The digital-era models abandoned the single sequence for a multi-dimensional profile: a firm can be advanced in technology and immature in talent, and it is precisely that asymmetry that predicts failure. Maturity thereby became a diagnostic instrument rather than a scorecard.

RELATED DEFINITIONS IN THIS COURSE

[Digital Transformation - Unit 1](#)
[Technology Adoption Lifecycle - Unit 1](#)
[AI Adoption - Unit 4](#)
[AI Roadmap - Unit 4](#)
[Organisational Culture - Unit 2](#)

SOURCES (HARVARD REFERENCING STYLE)

- Chanias, S. and Hess, T. (2016) 'How digital are we? Maturity models for the assessment of a company's status in the digital transformation', Management Report 2/2016. Munich: LMU Institute for Information Systems and New Media.
- Kane, G.C., Palmer, D., Phillips, A.N., Kiron, D. and Buckley, N. (2017) Achieving Digital Maturity. Cambridge, MA: MIT Sloan Management Review and Deloitte University Press.
- Nolan, R.L. (1973) 'Managing the computer resource: a stage hypothesis', Communications of the ACM, 16(7), pp. 399-405.
- Teichert, R. (2019) 'Digital transformation maturity: a systematic review of literature', Acta Universitatis Agriculturae et Silviculturae Mendelianae Brunensis, 67(6), pp. 1673-1687.

Digitisation and Digitalisation

GLOSSARY 01 - SHEET 4 / 8

Converting analogue to digital, versus changing how work is done with digital means

PROFESSIONAL DEFINITION

Digitisation is the technical conversion of analogue information into a digital format, leaving the underlying process unchanged. Digitalisation is the use of digital technologies to change a business process, business model or the way value is created and captured. The terms are not synonymous and not interchangeable, and the distinction between them separates cost-reduction projects from transformation (Gartner, 2019; Verhoef et al., 2021).

WHO DEFINED IT ORIGINALLY

The distinction was formalised for management research by Verhoef, Broekhuizen, Bart, Bhattacharya, Qi Dong, Fabian and Haenlein (2021) in the Journal of Business Research, building on Tilson, Lyytinen and Sorensen's (2010) information-systems account of digital infrastructure and on Gartner's practitioner glossary, which had already separated the two terms for its client base.

WHY THIS DEFINITION WAS DEVELOPED

Firms were reporting digital transformation while describing scanning projects. Because both activities consume the same budget line and use the same vocabulary, boards could not tell whether they were funding a change of medium or a change of business. Naming the two stages made the sequence explicit and made it possible to ask which one a given initiative actually was.

FIGURE 1 - THREE PHASES, THREE RETURNS

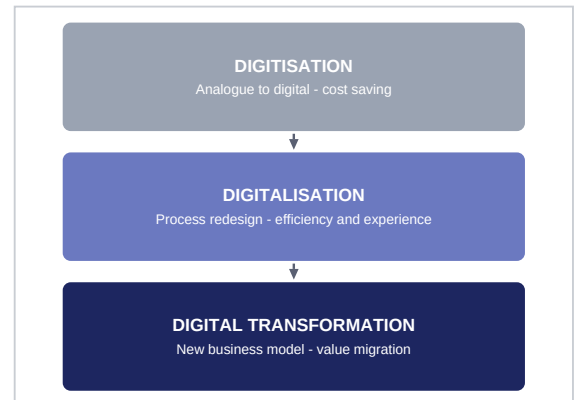


Figure 1. The three phases are sequential in capability but distinct in expected return. Attributing transformation-level benefits to a digitisation-level project is the most common source of overstated business cases (after Verhoef et al., 2021).

HOW IT IS USED

Classify each initiative as digitisation, digitalisation or transformation; expect digitisation to return cost savings only, and treat any claim of strategic return from a pure digitisation project as a category error.

WHEN IT IS USED

At portfolio review when benefits are being attributed, at business-case approval, and in post-implementation review when promised strategic benefits have not appeared.

BY WHOM IT IS USED

Programme managers classifying the initiative portfolio, finance business partners validating business cases, and transformation directors reporting benefits to the board.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor practice was a single undifferentiated term, digital, applied to everything from document scanning to platform business models. Verhoef and colleagues' three-phase framing supplies an ordered vocabulary in which each phase has its own expected return, its own risk profile and its own governance requirement, so that misattributed benefits become visible at the point of approval rather than at post-implementation review.

RELATED DEFINITIONS IN THIS COURSE

[Digital Transformation - Unit 1](#)
[Business Model Innovation - Unit 1](#)
[Digital Maturity - Unit 1](#)
[AI Operating Model - Unit 3](#)

SOURCES (HARVARD REFERENCING STYLE)

Gartner (2019) Gartner Glossary: Digitization, Digitalization. Stamford, CT: Gartner, Inc.

Tilson, D., Lyytinen, K. and Sorensen, C. (2010) 'Digital infrastructures: the missing IS research agenda', Information Systems Research, 21(4), pp. 748-759.

Verhoef, P.C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Qi Dong, J., Fabian, N. and Haenlein, M. (2021) 'Digital transformation: a multidisciplinary reflection and research agenda', Journal of Business Research, 122, pp. 889-901.

Business Model Innovation

Deliberate change to how a firm creates, delivers and captures value

GLOSSARY 01 - SHEET 5 / 8

PROFESSIONAL DEFINITION

Business model innovation is a deliberate and non-trivial change to one or more elements of the system by which a firm creates value for customers, delivers that value, and captures a share of it as revenue and margin. It is distinguished from product innovation because the object changed is the architecture of the business rather than the offering itself (Osterwalder and Pigneur, 2010; Amit and Zott, 2012).

WHO DEFINED IT ORIGINALLY

The canonical operational definition and its nine-block canvas were developed by Alexander Osterwalder in his 2004 doctoral thesis at HEC Lausanne under Yves Pigneur and published as *Business Model Generation* (2010). The strategic-theoretic account is Amit and Zott's (2001, 2012) activity-system perspective; Teece (2010) supplied the link to profit capture and dynamic capabilities.

WHY THIS DEFINITION WAS DEVELOPED

Strategy tools of the 1990s analysed industry position and competitive advantage but were silent on the mechanism by which a firm actually earned money. As digital entrants began competing on that mechanism rather than on the product, firms needed a shared, visual vocabulary for describing and redesigning it quickly enough to respond.

FIGURE 1 - CREATE, DELIVER, CAPTURE

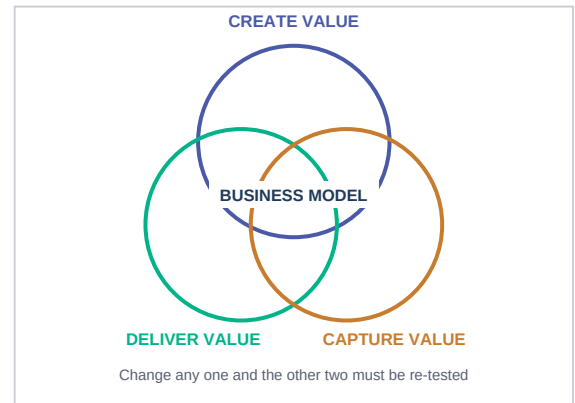


Figure 1. A business model is the intersection of three questions, not one of them. Innovation that changes value creation without changing capture produces adoption without margin - the recurring outcome of AI pilots priced as free internal services (after Osterwalder and Pigneur, 2010; Teece, 2010).

HOW IT IS USED

Map the current model across its nine blocks, identify which blocks a digital or AI-enabled competitor is attacking, and prototype alternative configurations before committing capital to any of them.

WHEN IT IS USED

When margins erode without loss of product quality, when a new entrant competes on a different revenue logic, at the start of a corporate venturing initiative, and when an AI capability makes a previously uneconomic customer segment viable.

BY WHOM IT IS USED

Strategy teams and corporate development, business unit general managers who own the profit and loss, and innovation and venturing units testing alternative configurations.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor instrument was the value chain (Porter, 1985), which decomposed a firm into sequential activities and located margin at the end of them. The business model canvas replaced the sequence with a system in which customer segments, channels, partners and revenue logic are altered simultaneously, and it made the object of change visual and testable. Where the value chain explained where cost sat, the canvas explains where the business could be different.

RELATED DEFINITIONS IN THIS COURSE

[Digital Transformation - Unit 1](#)
[Disruptive Innovation - Unit 1](#)
[Value Migration - Unit 1](#)
[Scalable AI Business Model - Unit 3](#)
[Profit Pool - Unit 3](#)

SOURCES (HARVARD REFERENCING STYLE)

- Amit, R. and Zott, C. (2012) 'Creating value through business model innovation', MIT Sloan Management Review, 53(3), pp. 41-49.
- Osterwalder, A. and Pigneur, Y. (2010) *Business Model Generation: A Handbook for Visionaries, Game Changers, and Challengers*. Hoboken, NJ: John Wiley & Sons.
- Porter, M.E. (1985) *Competitive Advantage: Creating and Sustaining Superior Performance*. New York: Free Press.
- Teece, D.J. (2010) 'Business models, business strategy and innovation', Long Range Planning, 43(2-3), pp. 172-194.

Disruptive Innovation

GLOSSARY 01 - SHEET 6 / 8

A process by which a low-end or new-market entrant displaces established competitors

PROFESSIONAL DEFINITION

Disruptive innovation is a process by which a product or service initially takes root in simple applications at the low end of a market, or creates a new market among non-consumers, and then moves upmarket over time until it displaces established competitors. The term describes the trajectory of an entrant relative to incumbent performance, not the novelty or impressiveness of the technology itself (Christensen, 1997; Christensen, Raynor and McDonald, 2015).

WHO DEFINED IT ORIGINALLY

Clayton M. Christensen developed the theory at Harvard Business School through study of the disk-drive industry, publishing it as *The Innovator's Dilemma* (1997). Christensen, Raynor and McDonald (2015) subsequently narrowed the definition in *Harvard Business Review* after two decades of misapplication. The critical literature, notably Lepore (2014) and King and Baatartogtokh (2015), disputes its predictive power.

WHY THIS DEFINITION WAS DEVELOPED

Christensen sought to explain why competently managed, customer-focused market leaders repeatedly lost to inferior entrants. His answer inverted the usual account: incumbents fail not through complacency but by correctly following their most profitable customers upmarket, which vacates the low end and makes the entrant's ascent possible.

FIGURE 1 - THE DISRUPTIVE TRAJECTORY

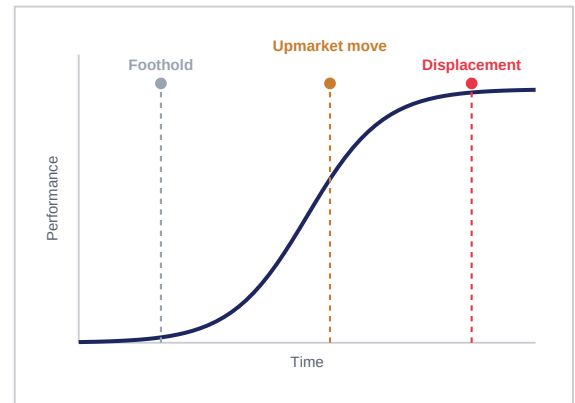


Figure 1. The entrant begins below the performance the mainstream market demands and improves faster than that demand rises. Displacement occurs when the two lines cross, by which point the incumbent's response options are largely exhausted (after Christensen, 1997).

HOW IT IS USED

Test whether a threat matches the trajectory - foothold among the over-served or non-consuming, initially inferior on mainstream attributes, improving faster than the market's requirements - and respond only if it does, because the response to disruption differs from the response to ordinary competition.

WHEN IT IS USED

In competitor analysis when an entrant is dismissed as too basic to matter, at capital allocation when an internal proposal targets a segment the firm currently declines to serve, and in board discussion of an entrant's growth rate.

BY WHOM IT IS USED

Strategy and corporate development teams assessing entrants, business unit leaders defending mature segments, and boards deciding whether to fund a business that would cannibalise existing revenue.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Its predecessor was the S-curve and technology-substitution literature (Foster, 1986), which located the cause of incumbent failure in a technical discontinuity that management failed to see. Christensen relocated the cause in the resource-allocation process: the incumbent sees the entrant clearly and rationally declines to serve its market. This shifted the remedy from better technology forecasting to organisational separation of the disruptive business, a materially different management prescription.

RELATED DEFINITIONS IN THIS COURSE

[Business Model Innovation - Unit 1](#)
[Value Migration - Unit 1](#)
[Technology Adoption Lifecycle - Unit 1](#)
[Competitive Advantage - Unit 3](#)
[General Purpose Technology - Unit 5](#)

SOURCES (HARVARD REFERENCING STYLE)

Christensen, C.M. (1997) *The Innovator's Dilemma: When New Technologies Cause Great Firms to Fail*. Boston, MA: Harvard Business School Press.
 Christensen, C.M., Raynor, M.E. and McDonald, R. (2015) 'What is disruptive innovation?', *Harvard Business Review*, 93(12), pp. 44-53.
 King, A.A. and Baatartogtokh, B. (2015) 'How useful is the theory of disruptive innovation?', *MIT Sloan Management Review*, 57(1), pp. 77-90.
 Lepore, J. (2014) 'The disruption machine', *The New Yorker*, 23 June.

Technology Adoption Lifecycle

GLOSSARY 01 - SHEET 7 / 8

The sequence of adopter groups through which an innovation diffuses

PROFESSIONAL DEFINITION

The technology adoption lifecycle is the sequence of distinct adopter categories - innovators, early adopters, early majority, late majority and laggards - through which an innovation diffuses over time, distributed approximately normally across a population. Each category adopts for different reasons and responds to different evidence, so a message that persuades one will not persuade the next (Rogers, 1962; Moore, 1991).

WHO DEFINED IT ORIGINALLY

Everett M. Rogers established the categories and their proportions in *Diffusion of Innovations* (1962), synthesising 508 diffusion studies across agriculture, medicine and education at Ohio State University. Geoffrey A. Moore (1991) added the discontinuity between early adopters and the early majority, which he named the chasm, from observation of Silicon Valley technology firms.

WHY THIS DEFINITION WAS DEVELOPED

Rogers sought a general theory of why demonstrably superior innovations spread slowly or not at all, having observed the delayed uptake of hybrid seed corn by Iowa farmers. Moore's later addition addressed a specific commercial failure pattern: products that succeeded with visionary buyers and then stalled, because the early majority required references from peers that the early adopters could not supply.

FIGURE 1 - THE ADOPTER DISTRIBUTION AND THE CHASM

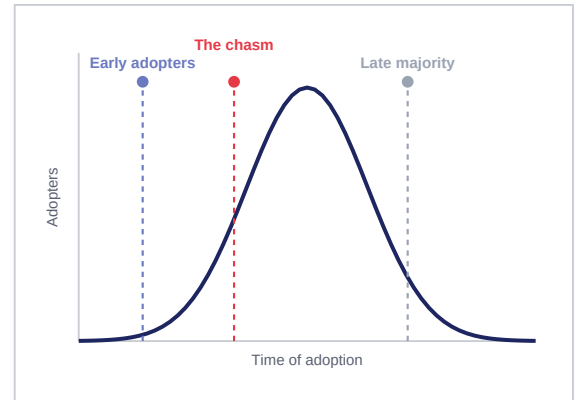


Figure 1. Adoption is approximately normally distributed across the population. Moore's chasm sits between the visionary early adopters and the pragmatic early majority; internal AI rollouts most often stall at exactly this point (after Rogers, 1962; Moore, 1991).

HOW IT IS USED

Identify which adopter group a given internal or external audience belongs to, and match the evidence offered to that group - novelty and advantage for early adopters, peer proof and low switching risk for the early majority.

WHEN IT IS USED

When planning the rollout sequence of an internal AI capability, in go-to-market planning for an AI-enabled product, and in diagnosing why a successful pilot has not spread beyond its original team.

BY WHOM IT IS USED

Change managers sequencing internal rollout, product and marketing leaders planning commercial launch, and adoption leads diagnosing stalled deployments.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Before Rogers, adoption was treated as a single aggregate rate to be accelerated by more information. Rogers replaced the aggregate with a segmented population having different decision criteria, which made targeted rather than uniform communication possible. Moore's contribution was to deny that the sequence is continuous: the gap between early adopters and the early majority is a discontinuity requiring a change of strategy, not merely more of the same effort.

RELATED DEFINITIONS IN THIS COURSE

[Digital Maturity - Unit 1](#)
[Disruptive Innovation - Unit 1](#)
[AI Adoption - Unit 4](#)
[Technology Acceptance Model - Unit 4](#)
[Change Management - Unit 4](#)

SOURCES (HARVARD REFERENCING STYLE)

- Moore, G.A. (1991) *Crossing the Chasm: Marketing and Selling High-Tech Products to Mainstream Customers*. New York: HarperBusiness.
- Rogers, E.M. (1962) *Diffusion of Innovations*. New York: Free Press.
- Rogers, E.M. (2003) *Diffusion of Innovations*. 5th edn. New York: Free Press.
- Ryan, B. and Gross, N.C. (1943) 'The diffusion of hybrid seed corn in two Iowa communities', *Rural Sociology*, 8(1), pp. 15-24.

Value Migration

GLOSSARY 01 - SHEET 8 / 8

The flow of economic value away from obsolete business designs toward designs that better serve priorities

PROFESSIONAL DEFINITION

Value migration is the movement of economic value away from business designs that no longer meet customer priorities and toward designs that do, observable as a sustained shift in market value relative to revenue across the competitors in an industry. The unit of analysis is the business design rather than the firm or the product, and migration can occur while the losing firm's revenue is still growing (Slywotzky, 1996).

WHO DEFINED IT ORIGINALLY

Adrian J. Slywotzky developed the concept at Mercer Management Consulting and published it as Value Migration (1996), following a study of value shifts across some thirty industries. The framing extends Drucker's (1994) theory of the business and connects to Christensen's contemporaneous work on incumbent displacement.

WHY THIS DEFINITION WAS DEVELOPED

Executives were monitoring revenue and market share and were surprised by collapses in market value that those measures had not signalled. Slywotzky argued that value moves before revenue does, so a construct anchored on business design and market-value-to-revenue ratios could give early warning that share-based metrics could not.

FIGURE 1 - THREE STATES OF A BUSINESS DESIGN

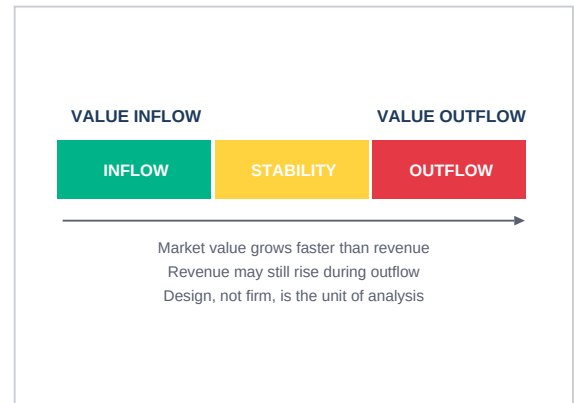


Figure 1. Every business design occupies one of three states. The diagnostic difficulty is that outflow begins while revenue is still growing, which is why share-based reporting detects it only after the reconfiguration options have narrowed (after Slywotzky, 1996).

HOW IT IS USED

Compute market value to revenue for each competitor over time, identify which designs are absorbing value and which are shedding it, and locate the firm's own design on that map before the shift reaches its revenue line.

WHEN IT IS USED

In annual strategic review, when an industry's profit pool is visibly reallocating, when a competitor's valuation diverges from its revenue trajectory, and when assessing whether an AI capability changes which design captures the margin.

BY WHOM IT IS USED

Corporate strategy and investor relations, chief financial officers explaining valuation divergence, and boards judging whether to defend or to reconfigure the current design.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor measures were market share and revenue growth, which are lagging and describe position within an existing design rather than the viability of the design itself. Value migration substitutes a leading indicator built on relative market value, and shifts the question from how much of this market do we hold to whether this market is still where the value is. Slywotzky's later work on profit patterns extended it into the profit-pool analysis used in Unit 3.

RELATED DEFINITIONS IN THIS COURSE

[Business Model Innovation - Unit 1](#)
[Disruptive Innovation - Unit 1](#)
[Profit Pool - Unit 3](#)
[Competitive Advantage - Unit 3](#)
[AI Strategy - Unit 3](#)

SOURCES (HARVARD REFERENCING STYLE)

- Drucker, P.F. (1994) 'The theory of the business', Harvard Business Review, 72(5), pp. 95-104.
- Slywotzky, A.J. (1996) Value Migration: How to Think Several Moves Ahead of the Competition. Boston, MA: Harvard Business School Press.
- Slywotzky, A.J. and Morrison, D.J. (1997) The Profit Zone: How Strategic Business Design Will Lead You to Tomorrow's Profits. New York: Times Business.

AI Literacy

The competence to use, interpret, evaluate and question AI systems in a working context

GLOSSARY 02 - SHEET 1 / 8

PROFESSIONAL DEFINITION

AI literacy is the set of competencies that enables a person to use artificial intelligence systems knowledgeably, to interpret and evaluate their outputs, to communicate about them accurately, and to judge when a system should not be relied upon. Since 2 February 2025 it carries a legal meaning in the European Union: Article 4 of the AI Act requires providers and deployers to ensure a sufficient level of AI literacy among staff operating AI systems on their behalf (Long and Magerko, 2020; Regulation (EU) 2024/1689, Article 4).

WHO DEFINED IT ORIGINALLY

The operational competency definition was proposed by Duri Long and Brian Magerko (2020) at the Georgia Institute of Technology, who synthesised seventeen competencies and fifteen design considerations from a review of the computational- and data-literacy literatures. Ng, Leung, Chu and Qiao (2021) extended it into an educational taxonomy. The legal obligation derives from Article 4 of Regulation (EU) 2024/1689.

WHY THIS DEFINITION WAS DEVELOPED

Organisations were responding to AI by training either everyone in prompting or a few specialists in modelling, and neither addressed the actual failure mode: managers who could not evaluate an output they had been given. A competency-based construct made it possible to specify what a non-technical employee must be able to do, and the EU subsequently made that specification a compliance duty rather than a development preference.

FIGURE 1 - FOUR LEVELS OF AI LITERACY

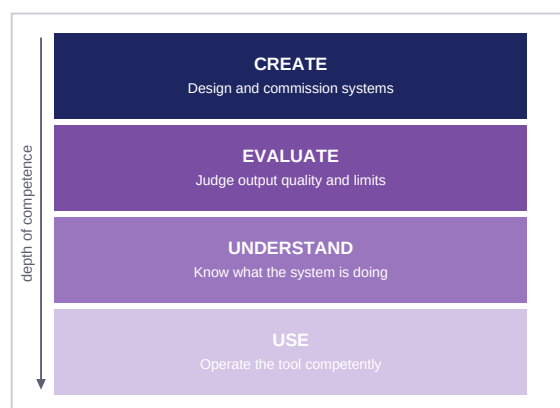


Figure 1. Literacy is cumulative. Most organisational programmes deliver the bottom layer and report the second; Article 4 of the AI Act is satisfied only where staff can perform the evaluation layer for the systems they operate (after Long and Magerko, 2020).

HOW IT IS USED

Map required competencies by role rather than by seniority, assess the current level against them, and close the gap with role-specific curricula; document the programme, since Article 4 compliance must be evidenced.

WHEN IT IS USED

Before granting staff access to an AI system, at onboarding, when a new class of system enters the organisation, and at the annual compliance review required under the AI Act.

BY WHOM IT IS USED

Human resources and learning functions designing the curriculum, line managers certifying role readiness, compliance officers evidencing Article 4, and every employee whose work involves an AI output.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor construct was digital literacy, which concerned the operation of tools whose behaviour was deterministic and inspectable. AI literacy adds the competencies that probabilistic and opaque systems require - recognising the limits of training data, interpreting confidence, identifying plausible-sounding error - and it locates the critical competence in evaluation rather than operation. Article 4 converted it from a capability model into an enforceable obligation, which no prior literacy construct had.

RELATED DEFINITIONS IN THIS COURSE

Artificial Intelligence - Unit 1

Human-in-the-Loop - Unit 2

Algorithm Aversion - Unit 2

Reskilling and Upskilling - Unit 4

EU AI Act - Unit 6

SOURCES (HARVARD REFERENCING STYLE)

European Parliament and Council (2024) Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence, Article 4. Official Journal of the European Union, L, 12 July 2024.

Long, D. and Magerko, B. (2020) 'What is AI literacy? Competencies and design considerations', in Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. New York: ACM, pp. 1-16.

Ng, D.T.K., Leung, J.K.L., Chu, S.K.W. and Qiao, M.S. (2021) 'Conceptualizing AI literacy: an exploratory review', Computers and Education: Artificial Intelligence, 2, 100041.

Psychological Safety

A shared belief that the team is safe for interpersonal risk-taking

GLOSSARY 02 - SHEET 2 / 8

PROFESSIONAL DEFINITION

Psychological safety is a shared belief held by members of a team that the team is safe for interpersonal risk-taking: that speaking up with a question, a concern, a mistake or a dissenting view will not lead to embarrassment, rejection or punishment. It is a property of the group climate rather than of individual personality, and it is distinct from trust, which concerns a dyad, and from cohesion, which can suppress dissent (Edmondson, 1999).

WHO DEFINED IT ORIGINALLY

Amy C. Edmondson defined and operationalised the construct at Harvard Business School in a 1999 study of 51 work teams in a manufacturing company, published in *Administrative Science Quarterly*. The antecedents are Schein and Bennis's (1965) work on organisational change and Kahn's (1990) study of personal engagement at work.

WHY THIS DEFINITION WAS DEVELOPED

Edmondson had found that better hospital units reported more medication errors, not fewer. Rather than accepting the perverse conclusion, she inferred that reporting rates measured climate rather than error rates, and needed a construct that explained why some teams surface problems and others conceal them. The construct then explained a general organisational phenomenon: the systematic suppression of upward bad news.

FIGURE 1 - SAFETY AND STANDARDS ARE INDEPENDENT

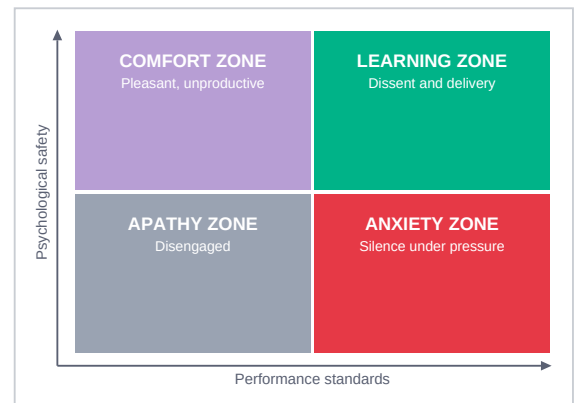


Figure 1. Psychological safety is not the absence of pressure. High standards without safety produce the anxiety zone, in which problems are known but unreported - the condition under which AI incidents accumulate silently (after Edmondson, 2018).

HOW IT IS USED

Measure the seven-item team-level scale, act on the result at team rather than individual level, and treat leader responses to bad news as the principal lever, since the climate is set by what happens the first time someone speaks up.

WHEN IT IS USED

Before an AI deployment where staff must report anomalies, after any incident that was known about before it occurred, at team formation, and whenever adoption metrics look uniformly positive.

BY WHOM IT IS USED

Team leaders whose responses set the climate, human resources functions running the diagnostic, AI governance leads who depend on incident reporting, and executives interpreting unanimously favourable pilot reports.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Its predecessor as an explanation of team performance was cohesion, which the group dynamics literature had shown could reduce performance by discouraging dissent - Janis's (1972) groupthink. Psychological safety separates the willingness to disagree from the desire to belong, and it is measured at the level of the team rather than the individual. This made it possible to explain why high-performing teams report more errors, an outcome that cohesion-based models predicted backwards.

RELATED DEFINITIONS IN THIS COURSE

Organisational Culture - Unit 2

Decision Rights - Unit 2

Change Management - Unit 4

AI Governance - Unit 6

Human-in-the-Loop - Unit 2

SOURCES (HARVARD REFERENCING STYLE)

- Edmondson, A.C. (1999) 'Psychological safety and learning behavior in work teams', *Administrative Science Quarterly*, 44(2), pp. 350-383.
- Edmondson, A.C. (2018) *The Fearless Organization: Creating Psychological Safety in the Workplace for Learning, Innovation, and Growth*. Hoboken, NJ: John Wiley & Sons.
- Kahn, W.A. (1990) 'Psychological conditions of personal engagement and disengagement at work', *Academy of Management Journal*, 33(4), pp. 692-724.
- Newman, A., Donohue, R. and Eva, N. (2017) 'Psychological safety: a systematic review of the literature', *Human Resource Management Review*, 27(3), pp. 521-535.

Upper Echelons Theory

GLOSSARY 02 - SHEET 3 / 8

Organisational outcomes are partly predicted by the characteristics of the top management team

PROFESSIONAL DEFINITION

Upper echelons theory holds that organisational outcomes - strategic choices and performance levels - are partially predicted by the observable and psychological characteristics of the dominant coalition of senior managers, because executives act on their personalised interpretations of the situations they face, and those interpretations are shaped by their experience, values and cognitive base (Hambrick and Mason, 1984; Hambrick, 2007).

WHO DEFINED IT ORIGINALLY

Donald C. Hambrick and Phyllis A. Mason set out the theory in the Academy of Management Review in 1984 while at Columbia University. Hambrick (2007) revisited and refined it, adding managerial discretion and executive job demands as moderating conditions. The intellectual foundation is the behavioural theory of the firm (Cyert and March, 1963) and March and Simon's bounded rationality.

WHY THIS DEFINITION WAS DEVELOPED

Strategy research of the period modelled the firm as a rational actor responding to industry structure, which left substantial variance in strategic choice unexplained between firms facing identical conditions. Hambrick and Mason proposed that the residual lay in who was deciding, and made the argument testable by using observable executive characteristics as proxies for cognition that could not be measured directly.

FIGURE 1 - FROM EXECUTIVE COGNITION TO OUTCOME

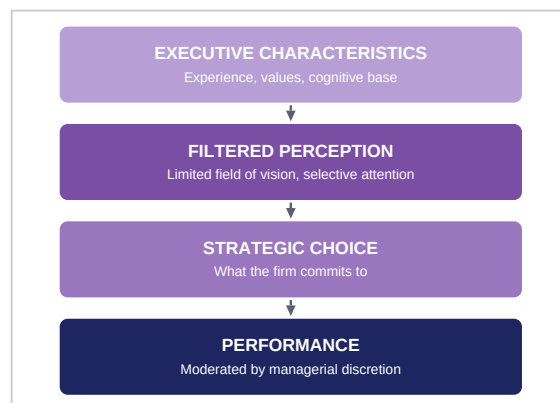


Figure 1. Executives do not respond to the situation but to their construction of it. The strength of the link from characteristics to outcome varies with managerial discretion, which is why the theory predicts well in loosely regulated markets and weakly in tightly constrained ones (after Hambrick, 2007).

HOW IT IS USED

Examine the composition, tenure, functional background and heterogeneity of the top team when explaining or predicting a strategic commitment; treat a homogeneous team as a predictor of narrow option generation, not merely of harmony.

WHEN IT IS USED

In board assessment of AI strategy proposals, at succession planning, when a firm's technology posture diverges sharply from its peers, and when diagnosing why an executive committee cannot agree what it decided.

BY WHOM IT IS USED

Boards and nomination committees composing the top team, chief executives assembling AI steering groups, and governance advisers assessing decision quality at the top.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor position was industrial-organisation strategy, in which management was a black box and structure determined conduct. Upper echelons opened the box while remaining empirically tractable, by substituting observable demographics for unobservable cognition. Hambrick's (2007) addition of managerial discretion resolved the theory's main weakness: executive characteristics predict outcomes strongly where discretion is high and weakly where it is constrained, which specifies the conditions of application rather than asserting universality.

RELATED DEFINITIONS IN THIS COURSE

[Decision Rights - Unit 2](#)
[Change Leadership - Unit 2](#)
[AI Strategy - Unit 3](#)
[Organisational Culture - Unit 2](#)
[Accountability Gap - Unit 6](#)

SOURCES (HARVARD REFERENCING STYLE)

Cyert, R.M. and March, J.G. (1963) A Behavioral Theory of the Firm. Englewood Cliffs, NJ: Prentice-Hall.

Hambrick, D.C. (2007) 'Upper echelons theory: an update', Academy of Management Review, 32(2), pp. 334-343.

Hambrick, D.C. and Mason, P.A. (1984) 'Upper echelons: the organization as a reflection of its top managers', Academy of Management Review, 9(2), pp. 193-206.

Organisational Culture

GLOSSARY 02 - SHEET 4 / 8

The pattern of shared basic assumptions a group has learned and teaches to new members

PROFESSIONAL DEFINITION

Organisational culture is the pattern of shared basic assumptions that a group learned as it solved its problems of external adaptation and internal integration, which has worked well enough to be considered valid and is therefore taught to new members as the correct way to perceive, think and feel in relation to those problems. It operates on three levels: visible artefacts, espoused values, and the underlying assumptions that actually govern behaviour (Schein, 1985).

WHO DEFINED IT ORIGINALLY

Edgar H. Schein developed the three-level model at the MIT Sloan School of Management and published it in *Organizational Culture and Leadership* (1985), drawing on clinical fieldwork rather than survey instruments. Competing operational traditions include Hofstede's (1980) dimensional measures and the Competing Values Framework of Cameron and Quinn (1999).

WHY THIS DEFINITION WAS DEVELOPED

Culture was being described through visible artefacts - offices, rituals, slogans - which changed readily and predicted nothing. Schein needed a model that explained why organisations revert after visible change, and located the answer one level down, in unexamined assumptions that members cannot articulate because they are not aware of holding them.

FIGURE 1 - THE THREE LEVELS OF CULTURE

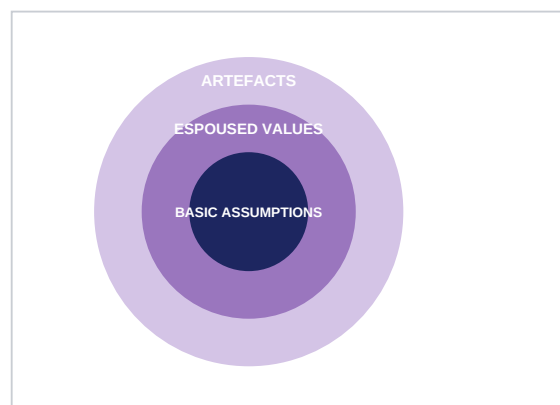


Figure 1. Only the outer ring is visible and only the inner ring is causal. Change programmes that alter artefacts and restate values while leaving the assumptions untouched produce reversion once attention moves elsewhere (after Schein, 1985).

HOW IT IS USED

Read the artefacts, elicit the espoused values, then look for the gap between what is espoused and what is done; the gap is where the operative assumption sits, and it is that assumption, not the value statement, that will govern behaviour under pressure.

WHEN IT IS USED

Before designing an AI adoption programme, during post-merger integration, when a stated value is contradicted by an observed decision, and when a change initiative has produced visible compliance without behavioural change.

BY WHOM IT IS USED

Executives sponsoring change, organisation development practitioners running the diagnosis, integration leads after acquisition, and adoption leads explaining why a technically sound system is not being used.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor tools were climate surveys, which measured how members currently felt in aggregate and were sensitive to recent events. Schein's model is structural rather than affective: it distinguishes what is displayed, what is claimed, and what is assumed, and locates causal force in the third level only. This explains a class of failure that climate instruments cannot - the organisation whose survey scores improve while its behaviour does not.

RELATED DEFINITIONS IN THIS COURSE

[Psychological Safety - Unit 2](#)
[Change Leadership - Unit 2](#)
[Change Management - Unit 4](#)
[Digital Maturity - Unit 1](#)
[AI Adoption - Unit 4](#)

SOURCES (HARVARD REFERENCING STYLE)

Cameron, K.S. and Quinn, R.E. (1999) *Diagnosing and Changing Organizational Culture*. Reading, MA: Addison-Wesley.
 Hofstede, G. (1980) *Culture's Consequences: International Differences in Work-Related Values*. Beverly Hills, CA: Sage.
 Schein, E.H. (1985) *Organizational Culture and Leadership*. San Francisco, CA: Jossey-Bass.
 Schein, E.H. and Schein, P. (2017) *Organizational Culture and Leadership*. 5th edn. Hoboken, NJ: John Wiley & Sons.

Change Leadership

The work of establishing direction and mobilising people through a transition

GLOSSARY 02 - SHEET 5 / 8

PROFESSIONAL DEFINITION

Change leadership is the work of establishing direction, aligning people to it and motivating them through a transition, as distinct from change management, which is the work of planning, resourcing, sequencing and controlling that transition. Leadership addresses the human transition that follows an organisational change; management addresses the change itself, and the two are complementary rather than interchangeable (Kotter, 1996; Bridges, 1991).

WHO DEFINED IT ORIGINALLY

John P. Kotter drew the leadership-management distinction at Harvard Business School and set out the eight-stage process in *Leading Change* (1996), following his 1995 Harvard Business Review analysis of why transformation efforts fail. William Bridges (1991) supplied the parallel psychological account of transition; Kurt Lewin's (1947) three-step model is the common antecedent.

WHY THIS DEFINITION WAS DEVELOPED

Kotter observed that most large-scale change efforts produced disappointing results despite competent project management, and concluded that firms were over-managing and under-leading. The distinction was needed because the two activities fail differently: management failure shows as a missed milestone, leadership failure shows as a delivered system that nobody uses.

FIGURE 1 - LEADERSHIP AND MANAGEMENT ARE DIFFERENT WORK

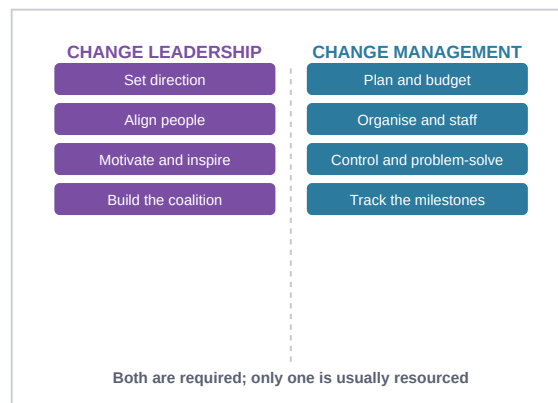


Figure 1. The two columns fail differently. Management failure appears as a missed milestone and is detected early; leadership failure appears as a completed deployment that nobody uses, and is detected only after the budget has been spent (after Kotter, 1996).

HOW IT IS USED

Use the eight stages as a diagnostic of what has been skipped rather than as a linear checklist; in practice most stalled programmes have skipped the first stage, urgency, and cannot be repaired at the stage where the symptom appeared.

WHEN IT IS USED

At the launch of an AI adoption programme, when a technically complete deployment has low usage, when resistance appears late, and when a coalition of sponsors has quietly dissolved.

BY WHOM IT IS USED

Executive sponsors of transformation, transformation directors, business unit leaders carrying the change into operations, and human resources partners supporting the transition.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Lewin's unfreeze-change-refreeze model treated organisations as systems returning to equilibrium, and gave no account of what leaders do inside each phase. Kotter's eight stages decompose the sequence into observable managerial acts and, critically, identify skipping as the dominant failure mode. Bridges added the insight that the transition runs on a different clock from the change: the organisation can be restructured on Monday while its people are still in the neutral zone months later.

RELATED DEFINITIONS IN THIS COURSE

Change Management - Unit 4

Organisational Culture - Unit 2

Psychological Safety - Unit 2

AI Adoption - Unit 4

Upper Echelons Theory - Unit 2

SOURCES (HARVARD REFERENCING STYLE)

- Bridges, W. (1991) *Managing Transitions: Making the Most of Change*. Reading, MA: Addison-Wesley.
- Kotter, J.P. (1995) 'Leading change: why transformation efforts fail', *Harvard Business Review*, 73(2), pp. 59-67.
- Kotter, J.P. (1996) *Leading Change*. Boston, MA: Harvard Business School Press.
- Lewin, K. (1947) 'Frontiers in group dynamics', *Human Relations*, 1(1), pp. 5-41.

Human-in-the-Loop

GLOSSARY 02 - SHEET 6 / 8

A control design in which a human decision is required within the system's operating cycle

PROFESSIONAL DEFINITION

Human-in-the-loop is a control design in which a human being is positioned inside the operating cycle of an automated system so that the system cannot complete a defined action without human input, review or approval. It is distinguished from human-on-the-loop, where the human supervises and may intervene but the system proceeds by default, and from human-in-command, where humans retain authority over whether the system operates at all (Amershi et al., 2014; HLEG AI, 2019).

WHO DEFINED IT ORIGINALLY

The design pattern originates in mid-twentieth-century control engineering and human-factors research, and was carried into machine learning through interactive machine learning, formalised by Amershi, Cakmak, Knox and Kulesza (2014). The three-way distinction used in European governance was set out by the High-Level Expert Group on AI in the Ethics Guidelines for Trustworthy AI (2019) and underpins the human oversight requirement in Article 14 of the AI Act.

WHY THIS DEFINITION WAS DEVELOPED

Early automation debates treated oversight as binary - automated or not - which gave designers no vocabulary for the intermediate arrangements that most real systems use. Naming three distinct positions made it possible to specify oversight at the level of the action rather than the system, and to require different arrangements for different risk levels.

FIGURE 1 - THREE POSITIONS OF HUMAN OVERSIGHT

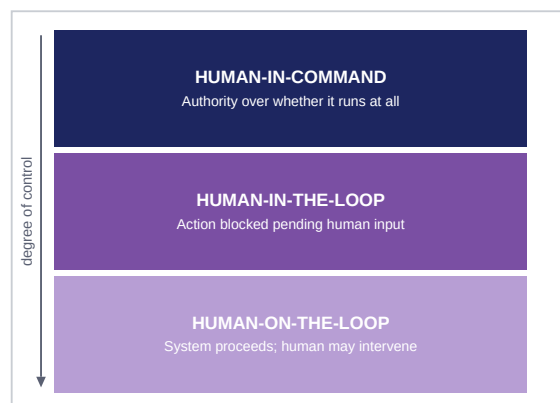


Figure 1. The three positions differ in what happens when the human does nothing. Only in the middle position does inaction stop the action, which is why it is the arrangement required for consequential decisions (after HLEG AI, 2019).

HOW IT IS USED

Specify, for each consequential action, which of the three positions applies, and verify that the human in that position has the information, the time and the authority to dissent; oversight without any of the three is nominal.

WHEN IT IS USED

At the design of any high-risk AI system, in conformity assessment under Article 14 of the AI Act, after an incident in which a human approved an output without examining it, and when automation bias is suspected.

BY WHOM IT IS USED

System designers and product owners specifying controls, risk and compliance functions evidencing oversight, operators occupying the loop, and auditors testing whether the control is real.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor concept was the manual override, a mechanism available in principle and unused in practice because the operator lacked the information or the time to exercise it. The in-the-loop, on-the-loop and in-command distinction relocates the design question from whether a human can intervene to whether the process requires them to, and pairs each position with an evidential burden. Article 14 then made that burden legally testable rather than a matter of design intent.

RELATED DEFINITIONS IN THIS COURSE

AI Literacy - Unit 2

Algorithm Aversion - Unit 2

Explainability - Unit 6

AI Governance - Unit 6

Accountability Gap - Unit 6

SOURCES (HARVARD REFERENCING STYLE)

Amershi, S., Cakmak, M., Knox, W.B. and Kulesza, T. (2014) 'Power to the people: the role of humans in interactive machine learning', AI Magazine, 35(4), pp. 105-120.

European Parliament and Council (2024) Regulation (EU) 2024/1689, Article 14 (human oversight). Official Journal of the European Union, L, 12 July 2024.

High-Level Expert Group on Artificial Intelligence (2019) Ethics Guidelines for Trustworthy AI. Brussels: European Commission.

Parasuraman, R. and Riley, V. (1997) 'Humans and automation: use, misuse, disuse, abuse', Human Factors, 39(2), pp. 230-253.

Algorithm Aversion

GLOSSARY 02 - SHEET 7 / 8

The tendency to discard an algorithm after seeing it err, even when it outperforms the human

PROFESSIONAL DEFINITION

Algorithm aversion is the tendency of people to lose confidence in an algorithmic forecaster more quickly than in a human forecaster after seeing each make the same mistake, and consequently to prefer the human even when the algorithm is demonstrably more accurate. Its counterpart, algorithm appreciation, is the tendency to over-rely on algorithmic advice in low-stakes or unfamiliar tasks; both are errors of calibration rather than of preference (Dietvorst, Simmons and Massey, 2015; Logg, Minson and Moore, 2019).

WHO DEFINED IT ORIGINALLY

Berkeley J. Dietvorst, Joseph P. Simmons and Cade Massey demonstrated and named the effect at the Wharton School in a 2015 Journal of Experimental Psychology: General paper across five experiments. Jennifer M. Logg, Julia A. Minson and Don A. Moore (2019) established the opposing appreciation effect. The lineage runs to Meehl's (1954) demonstration that statistical prediction outperforms clinical judgement.

WHY THIS DEFINITION WAS DEVELOPED

Meehl's finding had been replicated for six decades without changing practice, which suggested the obstacle was behavioural rather than evidential. Dietvorst and colleagues needed to identify the specific mechanism, and found it in asymmetric updating: an algorithmic error is treated as evidence of a defective system, a human error as an isolated lapse.

FIGURE 1 - ASYMMETRIC RESPONSE TO IDENTICAL ERROR

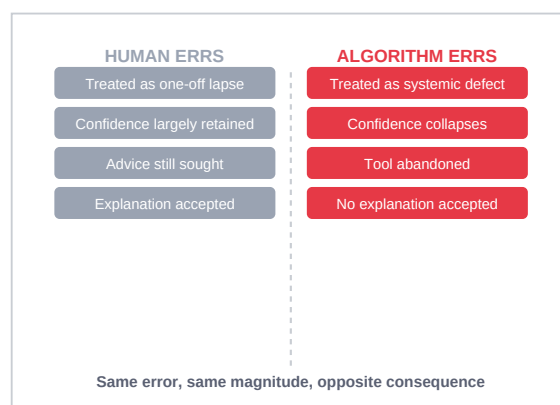


Figure 1. Aversion is an updating asymmetry, not a preference for humans. Granting users even a small right to adjust the model's output substantially restores usage, which makes the remedy a design decision (after Dietvorst, Simmons and Massey, 2015, 2018).

HOW IT IS USED

Anticipate the asymmetry in deployment design: give users limited ability to adjust the model's output, which the same authors showed substantially restores use without materially degrading accuracy, and report model performance against the human baseline rather than against perfection.

WHEN IT IS USED

When a technically superior model is being rejected by its intended users, after a visible model error, when designing the interface through which recommendations are delivered, and when adoption falls after go-live.

BY WHOM IT IS USED

Product owners designing decision-support interfaces, adoption leads diagnosing rejection, model risk functions setting acceptable error tolerances, and managers whose teams have stopped using a deployed tool.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor framing was resistance to change or lack of trust, which treated rejection as a general disposition to be addressed by training and communication. Algorithm aversion identifies a specific, testable mechanism triggered by a specific event - the observed error - and yields a design remedy rather than a persuasion remedy. Dietvorst, Simmons and Massey's (2018) follow-up showing that even trivial adjustment rights restore usage converted a behavioural finding into an interface specification.

RELATED DEFINITIONS IN THIS COURSE

Human-in-the-Loop - Unit 2

AI Literacy - Unit 2

Technology Acceptance Model - Unit 4

Explainability - Unit 6

AI Adoption - Unit 4

SOURCES (HARVARD REFERENCING STYLE)

- Dietvorst, B.J., Simmons, J.P. and Massey, C. (2015) 'Algorithm aversion: people erroneously avoid algorithms after seeing them err', Journal of Experimental Psychology: General, 144(1), pp. 114-126.
- Dietvorst, B.J., Simmons, J.P. and Massey, C. (2018) 'Overcoming algorithm aversion: people will use imperfect algorithms if they can (even slightly) modify them', Management Science, 64(3), pp. 1155-1170.
- Logg, J.M., Minson, J.A. and Moore, D.A. (2019) 'Algorithm appreciation: people prefer algorithmic to human judgment', Organizational Behavior and Human Decision Processes, 151, pp. 90-103.
- Meehl, P.E. (1954) Clinical versus Statistical Prediction. Minneapolis, MN: University of Minnesota Press.

Decision Rights

GLOSSARY 02 - SHEET 8 / 8

The formal allocation of who may decide, who must be consulted, and who is accountable

PROFESSIONAL DEFINITION

Decision rights are the formal allocation across roles of the authority to make a specified class of decision, the obligation to provide input to it, and the accountability for its consequences. Allocating them requires answering four questions for each decision: what is being decided, who decides, how the decision is made, and when it is made. Ambiguity in this allocation, rather than in the analysis, is the common cause of decisions that are made repeatedly and never implemented (Rogers and Blenko, 2006).

WHO DEFINED IT ORIGINALLY

The practitioner instrument in general use, RAPID, was developed by Paul Rogers and Marcia Blenko at Bain & Company and published in Harvard Business Review in 2006. Its theoretical foundation is Jensen and Meckling's (1992) treatment of specific knowledge and the co-location of decision rights with the knowledge required to exercise them, itself grounded in Hayek's account of dispersed knowledge.

WHY THIS DEFINITION WAS DEVELOPED

Firms were responding to slow decision-making by restructuring, which changed reporting lines without changing who could actually decide. Rogers and Blenko argued that the binding constraint was role ambiguity, and that it could be resolved decision by decision without any change to the organisation chart - a materially cheaper intervention.

FIGURE 1 - FIVE ROLES IN ONE DECISION

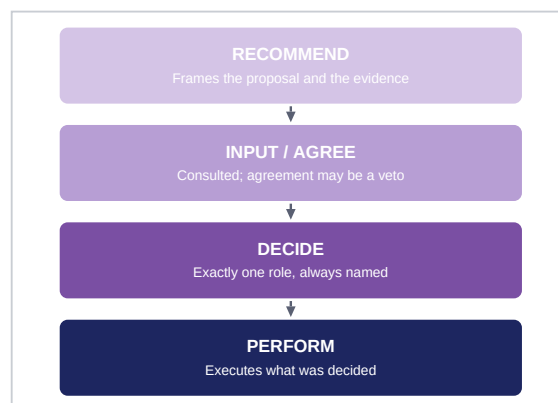


Figure 1. The decide role is singular by construction. Where two roles believe they hold it, the decision is made repeatedly and implemented never - the pattern most often mistaken for analytical disagreement (after Rogers and Blenko, 2006).

HOW IT IS USED

Take the specific decision, not the function; name the single role that decides and the roles that recommend, agree, provide input and perform; publish the allocation, since an unpublished allocation is indistinguishable from none.

WHEN IT IS USED

When an AI investment decision has been revisited three times, at the formation of an AI steering committee, at the boundary between central and business unit authority over models, and whenever accountability for a deployed system cannot be named.

BY WHOM IT IS USED

Executive committees allocating authority over AI investment, governance leads defining model approval paths, business unit heads negotiating central-local boundaries, and programme offices resolving decision deadlock.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Its predecessor was the RACI matrix, which assigned responsibility for tasks within a project and treated the decision as a task like any other. RAPID separates the authority to decide from the obligation to agree, and makes the decider singular by construction, which removes the deadlock that RACI's multiple accountable parties permit. Applying it to the decision rather than the process is what allows the intervention to be made without restructuring.

RELATED DEFINITIONS IN THIS COURSE

[Upper Echelons Theory - Unit 2](#)
[AI Governance - Unit 6](#)
[Accountability Gap - Unit 6](#)
[AI Operating Model - Unit 3](#)
[Psychological Safety - Unit 2](#)

SOURCES (HARVARD REFERENCING STYLE)

- Jensen, M.C. and Meckling, W.H. (1992) 'Specific and general knowledge and organizational structure', in Werin, L. and Wijkander, H. (eds) Contract Economics. Oxford: Blackwell, pp. 251-274.
- Rogers, P. and Blenko, M. (2006) 'Who has the D? How clear decision roles enhance organizational performance', Harvard Business Review, 84(1), pp. 52-61.
- Blenko, M.W., Mankins, M.C. and Rogers, P. (2010) Decide and Deliver: Five Steps to Breakthrough Performance in Your Organization. Boston, MA: Harvard Business Review Press.

AI Strategy

A coherent set of choices about where AI will be applied and what will not be done

GLOSSARY 03 - SHEET 1 / 8

PROFESSIONAL DEFINITION

An AI strategy is a coherent set of integrated choices specifying where in the business artificial intelligence will be applied, what capability will be built or acquired to apply it, how the resulting advantage will be defended, and - definitionally - what the firm will therefore stop doing. A list of intended applications without the corresponding renunciations is a budget, not a strategy, because it makes no choice (Rumelt, 2011; Iansiti and Lakhani, 2020).

WHO DEFINED IT ORIGINALLY

The requirement that strategy consist of choices with foregone alternatives is Porter's (1996), restated as the kernel of diagnosis, guiding policy and coherent action by Richard P. Rumelt (2011) at UCLA Anderson. The AI-specific application is developed by Marco Iansiti and Karim R. Lakhani (2020) at Harvard Business School through the concept of the AI factory and the operating-model constraint on scale.

WHY THIS DEFINITION WAS DEVELOPED

Firms were producing documents titled AI strategy that enumerated use cases without prioritising among them, which guaranteed that scarce data-engineering and change capacity would be spread across a portfolio too wide to complete any element of it. Rumelt's kernel supplies the test: if nothing was given up, no choice was made and no capacity was concentrated.

HOW IT IS USED

State the diagnosis of the competitive problem, the guiding policy for addressing it, and the coherent actions that follow; then list explicitly the initiatives and markets the firm will cease to pursue in order to fund them.

WHEN IT IS USED

At annual strategic planning, when the AI initiative portfolio exceeds delivery capacity, when a competitor's AI-enabled offering changes the basis of competition, and at board approval of multi-year AI investment.

BY WHOM IT IS USED

Chief executives and boards, chief strategy officers, chief data and AI officers translating strategy into capability, and business unit leaders whose plans must be reconciled with the group choice.

FIGURE 1 - THE KERNEL OF AN AI STRATEGY



Figure 1. The fourth block is the test. A document containing the first three and omitting the fourth has allocated no capacity and has therefore made no choice; it will be executed at the rate of the organisation's spare capacity, which is nil (after Rumelt, 2011).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor artefact was the technology roadmap, which sequenced planned capability without reference to competitive position or to what would be surrendered to fund it. Applying Rumelt's kernel converts a roadmap into a strategy by requiring a diagnosis - a claim about what is actually wrong - and by making renunciation a formal element rather than an implicit consequence. Iansiti and Lakhani's contribution is the constraint: the binding limit on AI advantage is the operating model, not the algorithm.

RELATED DEFINITIONS IN THIS COURSE

[Data Strategy - Unit 3](#)
[Competitive Advantage - Unit 3](#)
[AI Operating Model - Unit 3](#)
[Value Migration - Unit 1](#)
[AI Roadmap - Unit 4](#)

SOURCES (HARVARD REFERENCING STYLE)

Iansiti, M. and Lakhani, K.R. (2020) Competing in the Age of AI: Strategy and Leadership When Algorithms and Networks Run the World. Boston, MA: Harvard Business Review Press.

Porter, M.E. (1996) 'What is strategy?', Harvard Business Review, 74(6), pp. 61-78.

Rumelt, R.P. (2011) Good Strategy / Bad Strategy: The Difference and Why It Matters. New York: Crown Business.

Data Strategy

GLOSSARY 03 - SHEET 2 / 8

The choices governing how data is sourced, governed and used to serve the business strategy

PROFESSIONAL DEFINITION

A data strategy is the set of choices governing how an organisation sources, stores, governs, shares and exploits data in service of its business strategy. A central choice is the balance between defensive objectives - security, integrity, regulatory compliance - and offensive objectives - analytics, customer insight, competitive differentiation - since the architectures that optimise for one impede the other (DalleMule and Davenport, 2017).

WHO DEFINED IT ORIGINALLY

The offence-defence framing was set out by Leandro DalleMule, then chief data officer at AIG, and Thomas H. Davenport of Babson College in the Harvard Business Review in 2017. The single-source-of-truth versus multiple-versions-of-the-truth architecture that operationalises it is drawn from the data-warehousing tradition of Inmon (1992) and Kimball (1996).

WHY THIS DEFINITION WAS DEVELOPED

Chief data officers were being given two mandates that pull in opposite directions - control the data and monetise it - and were failing at both by pursuing them with one architecture. Naming the trade-off allowed the balance to be set deliberately, by industry and by regulatory exposure, rather than resolved by whichever function shouted last.

FIGURE 1 - THE OFFENCE-DEFENCE CONTINUUM

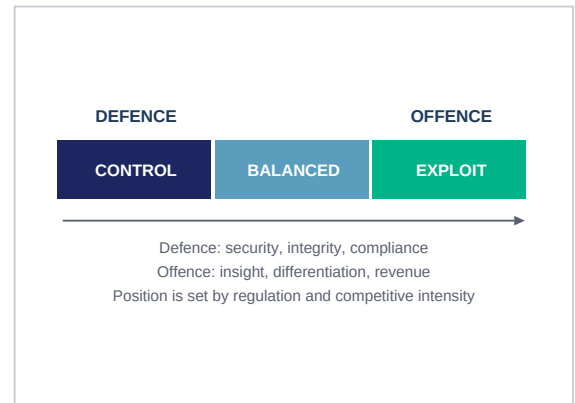


Figure 1. Every firm sits somewhere on this line and no firm sits at both ends. Architectures optimised for a single source of truth serve defence; architectures permitting multiple transformed versions serve offence (after DalleMule and Davenport, 2017).

HOW IT IS USED

Place the firm on the offence-defence continuum according to regulatory exposure and competitive intensity, then design a single source of truth for governed data while permitting business functions their own transformed versions above it.

WHEN IT IS USED

At the appointment of a chief data officer, before an AI programme that depends on data the firm does not yet control, at regulatory change, and when analytics teams are recreating the same dataset independently.

BY WHOM IT IS USED

Chief data officers setting the architecture, chief information security officers on the defensive side, analytics and AI leaders on the offensive side, and the executive committee arbitrating between them.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor was data governance framed purely as a control function, in which every request was assessed for risk and value creation had no advocate. The offence-defence continuum makes the trade-off explicit and positional: a regulated bank and a consumer platform should sit at different points, and each can then justify its architecture against its own position rather than against a universal ideal of control.

RELATED DEFINITIONS IN THIS COURSE

[AI Strategy - Unit 3](#)
[AI Operating Model - Unit 3](#)
[MLOps - Unit 4](#)
[Data Protection by Design - Unit 6](#)
[Competitive Advantage - Unit 3](#)

SOURCES (HARVARD REFERENCING STYLE)

DalleMule, L. and Davenport, T.H. (2017) 'What's your data strategy?', Harvard Business Review, 95(3), pp. 112-121.

Inmon, W.H. (1992) Building the Data Warehouse. New York: John Wiley & Sons.

Kimball, R. (1996) The Data Warehouse Toolkit. New York: John Wiley & Sons.

Davenport, T.H. and Bean, R. (2023) 'Data and AI leadership executive survey 2023', MIT Sloan Management Review, January.

Competitive Advantage

Superior value creation that rivals cannot readily replicate or substitute

GLOSSARY 03 - SHEET 3 / 8

PROFESSIONAL DEFINITION

A competitive advantage exists where a firm creates greater economic value than its rivals and the source of that value is not simultaneously being exploited by them. It is sustained where the underlying resource is valuable, rare, imperfectly imitable and organisationally exploitable - the VRIO conditions - and it is competitive parity, not advantage, where a resource is valuable but widely available (Barney, 1991; Porter, 1985).

WHO DEFINED IT ORIGINALLY

Jay B. Barney formalised the resource-based conditions at Texas A&M University in a 1991 Journal of Management article, building on Wernerfelt's (1984) resource-based view and Penrose's (1959) theory of the growth of the firm. Michael E. Porter's (1985) positioning account, in which advantage derives from activity configuration rather than resource endowment, remains the principal alternative.

WHY THIS DEFINITION WAS DEVELOPED

Industry-structure analysis explained why some industries were more profitable than others but not why firms within the same industry differed persistently. Barney's conditions relocated the explanation inside the firm and, crucially, specified the tests a claimed advantage must pass - which made most claimed advantages fail.

HOW IT IS USED

Apply the four tests in order to the specific resource claimed; a purchasable model or a commercially available dataset fails rarity immediately, which redirects attention to the proprietary data, workflow integration and complementary assets that may not.

WHEN IT IS USED

When assessing whether an AI capability confers advantage or merely restores parity, at investment approval for a differentiating capability, and when a competitor announces the same technology.

BY WHOM IT IS USED

Strategy teams testing advantage claims, investment committees, chief data and AI officers distinguishing proprietary from purchasable capability, and boards assessing the durability of a technology-based position.

FIGURE 1 - THE FOUR TESTS OF A RESOURCE

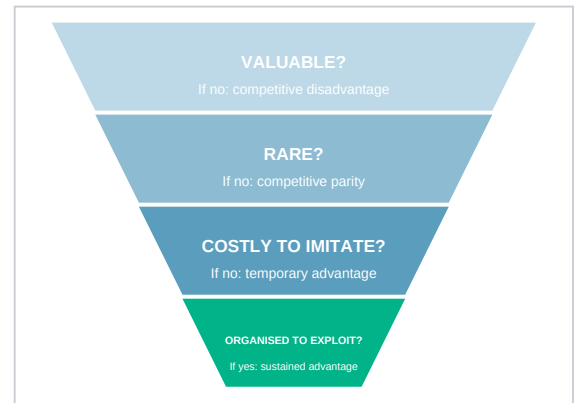


Figure 1. Each test removes candidates. A commercially available model typically fails the second test, which is why claimed AI advantage should be assessed against the proprietary data and workflow integration surrounding the model rather than the model itself (after Barney, 1991).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Porter's five forces located profitability in industry structure and treated firms within an industry as broadly similar. The resource-based view explains intra-industry variance, and VRIO converts the view into a sequence of falsifiable tests rather than a description. For AI specifically this is decisive: the model is rarely rare, so the four tests move the analysis to data, integration and organisational capability, where the durable difference actually sits.

RELATED DEFINITIONS IN THIS COURSE

[AI Strategy - Unit 3](#)
[Value Migration - Unit 1](#)
[Profit Pool - Unit 3](#)
[Build-Buy-Partner Decision - Unit 3](#)
[Disruptive Innovation - Unit 1](#)

SOURCES (HARVARD REFERENCING STYLE)

Barney, J. (1991) 'Firm resources and sustained competitive advantage', Journal of Management, 17(1), pp. 99-120.
 Penrose, E.T. (1959) The Theory of the Growth of the Firm. Oxford: Basil Blackwell.
 Porter, M.E. (1985) Competitive Advantage: Creating and Sustaining Superior Performance. New York: Free Press.
 Wernerfelt, B. (1984) 'A resource-based view of the firm', Strategic Management Journal, 5(2), pp. 171-180.

AI Operating Model

GLOSSARY 03 - SHEET 4 / 8

How AI capability, decision rights and delivery are organised across centre and business

PROFESSIONAL DEFINITION

An AI operating model is the arrangement of people, decision rights, processes, platform and governance through which an organisation converts AI capability into operating results. Its principal design choice is the distribution of capability between a central function and the business units - centralised, federated with a centre of excellence, or fully devolved - and each option trades delivery speed against consistency, reuse and control (Fountaine, McCarthy and Saleh, 2019).

WHO DEFINED IT ORIGINALLY

The three-model taxonomy in general management use was articulated by Tim Fountaine, Brian McCarthy and Tamim Saleh of McKinsey & Company in the Harvard Business Review in 2019. Its structural foundation is Galbraith's (1973) information-processing view of organisation design and the ambidexterity literature of Tushman and O'Reilly (1996).

WHY THIS DEFINITION WAS DEVELOPED

Firms were placing data scientists either entirely in a central laboratory, where they built work the business would not adopt, or entirely in business units, where they rebuilt the same infrastructure repeatedly. The taxonomy named a third arrangement and, more usefully, specified which conditions favour each, converting an ideological argument into a design decision.

FIGURE 1 - THREE OPERATING MODELS

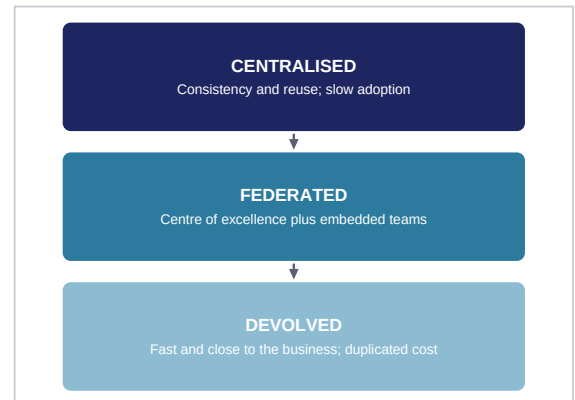


Figure 1. The three arrangements are not a maturity ladder to be climbed in one direction. Most organisations begin centralised to build the platform and move to federated as capability spreads; failures occur at the interfaces between centre and unit, not within either (after Fountaine, McCarthy and Saleh, 2019).

HOW IT IS USED

Select the model that matches AI maturity and business-unit heterogeneity, then specify the interfaces explicitly - who owns the platform, who owns the model, who owns the decision to deploy - because the failures occur at the interfaces rather than inside the units.

WHEN IT IS USED

When establishing an AI function, when a central team's output is not being adopted, at post-merger integration of two analytics organisations, and when platform costs are being duplicated across units.

BY WHOM IT IS USED

Chief data and AI officers designing the function, chief operating officers owning the interfaces, business unit leaders receiving the capability, and human resources planning the talent model.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor was the analytics centre of excellence, presented as a universal solution. Fountaine and colleagues treat the choice as contingent on maturity and heterogeneity rather than as a single best practice, and add the explicit prescription that the model should change as maturity rises - typically from centralised to federated. This makes the operating model a variable to be managed rather than a structure to be installed once.

RELATED DEFINITIONS IN THIS COURSE

[AI Strategy - Unit 3](#)
[Data Strategy - Unit 3](#)
[Decision Rights - Unit 2](#)
[MLOps - Unit 4](#)
[AI Governance - Unit 6](#)

SOURCES (HARVARD REFERENCING STYLE)

- Fountaine, T., McCarthy, B. and Saleh, T. (2019) 'Building the AI-powered organization', Harvard Business Review, 97(4), pp. 62-73.
 Galbraith, J.R. (1973) Designing Complex Organizations. Reading, MA: Addison-Wesley.
 Tushman, M.L. and O'Reilly, C.A. (1996) 'Ambidextrous organizations: managing evolutionary and revolutionary change', California Management Review, 38(4), pp. 8-30.

Build-Buy-Partner Decision

GLOSSARY 03 - SHEET 5 / 8

Choosing whether to develop internally, procure, or ally for a required capability

PROFESSIONAL DEFINITION

The build-buy-partner decision is the choice of governance mode for acquiring a required capability: internal development, market procurement, or a collaborative arrangement such as an alliance or joint venture. In transaction cost economics the choice turns on asset specificity, uncertainty and the frequency of the transaction, with internalisation preferred as specificity and uncertainty rise, because market contracting then becomes vulnerable to opportunism (Williamson, 1985; Capron and Mitchell, 2012).

WHO DEFINED IT ORIGINALLY

The theoretical basis is Oliver E. Williamson's transaction cost economics, developed from Ronald Coase's (1937) question about the nature of the firm and set out in *The Economic Institutions of Capitalism* (1985), for which Williamson shared the 2009 Nobel Memorial Prize. Laurence Capron and Will Mitchell (2012) developed the managerial decision framework in *Build, Borrow, or Buy*.

WHY THIS DEFINITION WAS DEVELOPED

Capability acquisition decisions were being taken on cost comparison alone, which systematically favoured buying because it ignored the costs of contracting, monitoring and eventual lock-in. Williamson's framework prices those costs; Capron and Mitchell added the observation that firms have a persistent default mode and apply it regardless of the transaction's characteristics.

FIGURE 1 - GOVERNANCE MODE BY ASSET SPECIFICITY

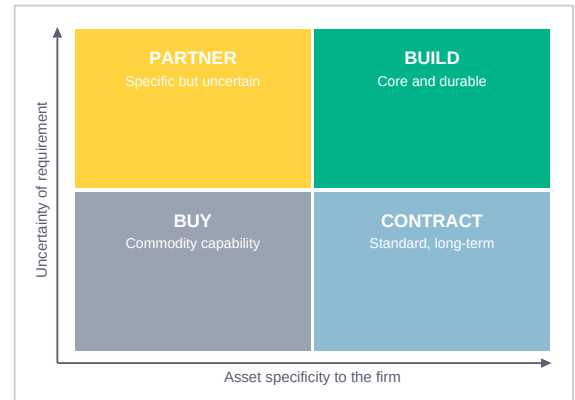


Figure 1. The mode follows the transaction, not the firm's habit. Foundation models typically sit in the buy quadrant while the data pipelines and workflow integration around them sit in the build quadrant, which is why the two are procured differently (after Williamson, 1985; Capron and Mitchell, 2012).

HOW IT IS USED

Assess the capability's specificity to the firm, the uncertainty around its future requirements and how often it will be transacted; internalise where all three are high, procure where all three are low, and use a partnership where they diverge.

WHEN IT IS USED

At foundation-model and AI platform selection, when deciding whether to hire or contract scarce AI talent, before signing a multi-year vendor commitment, and when reassessing an existing vendor relationship at renewal.

BY WHOM IT IS USED

Technology and procurement leaders, chief financial officers evaluating total cost of ownership, corporate development teams structuring alliances, and legal counsel assessing lock-in and exit.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor decision rule was make-or-buy on unit cost, a two-option comparison of production cost that ignored transaction cost entirely. Williamson added the third element - the cost of the transaction itself - which explains why firms internalise activities they could purchase more cheaply. Capron and Mitchell's contribution is behavioural: they show that the dominant error is not choosing wrongly on the merits but applying the firm's habitual mode without assessing them.

RELATED DEFINITIONS IN THIS COURSE

[AI Strategy - Unit 3](#)
[Competitive Advantage - Unit 3](#)
[AI Operating Model - Unit 3](#)
[Scalable AI Business Model - Unit 3](#)
[MLOps - Unit 4](#)

SOURCES (HARVARD REFERENCING STYLE)

Capron, L. and Mitchell, W. (2012) *Build, Borrow, or Buy: Solving the Growth Dilemma*. Boston, MA: Harvard Business Review Press.
 Coase, R.H. (1937) 'The nature of the firm', *Economica*, 4(16), pp. 386-405.
 Williamson, O.E. (1985) *The Economic Institutions of Capitalism*. New York: Free Press.

Scalable AI Business Model

GLOSSARY 03 - SHEET 6 / 8

A model whose marginal cost of serving an additional user does not rise with volume

PROFESSIONAL DEFINITION

A scalable AI business model is one in which the cost of serving an additional customer or processing an additional transaction does not rise in proportion to volume, so that value delivered grows faster than the resources consumed. Iansiti and Lakhani identify three properties that distinguish such models - the absence of operational diseconomies of scale, positive returns to scope from reusing data across domains, and continuous learning effects - and locate the constraint on all three in the operating model rather than the algorithm (Iansiti and Lakhani, 2020).

WHO DEFINED IT ORIGINALLY

Marco Iansiti and Karim R. Lakhani of Harvard Business School developed the account in *Competing in the Age of AI* (2020), extending their earlier work on digital platforms and drawing on Arthur's (1996) analysis of increasing returns and on Parker, Van Alstyne and Choudary's (2016) platform economics.

WHY THIS DEFINITION WAS DEVELOPED

Traditional firms scale by adding people, and their operating models therefore encounter diseconomies as coordination costs rise. Iansiti and Lakhani needed to explain why digitally native competitors did not, and located the difference in an architecture where the software, not the workforce, absorbs additional volume - which reframes the strategic question as one of operating-model design.

FIGURE 1 - THREE PROPERTIES OF A SCALABLE MODEL

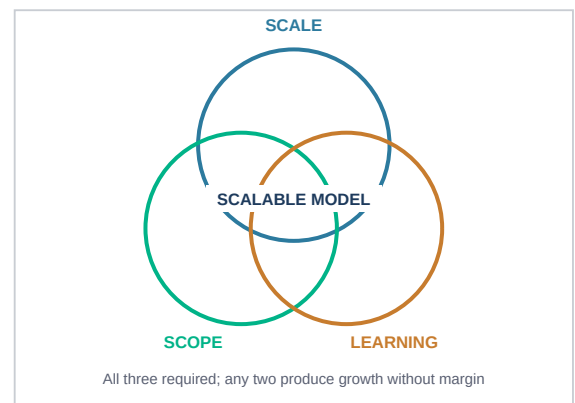


Figure 1. Scale without learning yields a large business that does not improve; learning without scale yields an improving business that cannot grow. The binding constraint on all three is the operating model, not the algorithm (after Iansiti and Lakhani, 2020).

HOW IT IS USED

Test each of the three properties against the specific model: whether marginal delivery cost is flat, whether the data collected is reusable in adjacent domains, and whether use improves the system; a model failing all three will not scale regardless of the technology used.

WHEN IT IS USED

At business-case approval for an AI-enabled product, when unit economics deteriorate as volume rises, when assessing a competitor's growth trajectory, and when a services business proposes an AI product.

BY WHOM IT IS USED

Product and business-model owners, chief financial officers testing unit economics, venturing and corporate development teams, and boards assessing whether growth will be profitable.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor concept was economies of scale from manufacturing economics, which assumed a fixed asset base amortised over volume and produced diminishing returns beyond an optimum. The AI-era account substitutes learning effects and scope economies, in which the asset improves rather than depreciates with use. Its important qualification is that these properties belong to the operating model: a firm can hold excellent models and still fail to scale if humans remain in the delivery path.

RELATED DEFINITIONS IN THIS COURSE

Business Model Innovation - Unit 1

AI Operating Model - Unit 3

Profit Pool - Unit 3

Competitive Advantage - Unit 3

Value Migration - Unit 1

SOURCES (HARVARD REFERENCING STYLE)

Arthur, W.B. (1996) 'Increasing returns and the new world of business', *Harvard Business Review*, 74(4), pp. 100-109.

Iansiti, M. and Lakhani, K.R. (2020) *Competing in the Age of AI: Strategy and Leadership When Algorithms and Networks Run the World*. Boston, MA: Harvard Business Review Press.

Parker, G.G., Van Alstyne, M.W. and Choudary, S.P. (2016) *Platform Revolution*. New York: W.W. Norton.

Profit Pool

GLOSSARY 03 - SHEET 7 / 8

The total profit earned across all activities in a value chain, mapped by segment

PROFESSIONAL DEFINITION

A profit pool is the total profit earned at all points along the value chain of an industry, disaggregated by activity or segment and displayed against the revenue share of each, so that the concentration of profit becomes visible independently of the concentration of revenue. Profit pools are frequently distributed very differently from revenue pools, and they reallocate over time as technology alters which activity is scarce (Gadiesh and Gilbert, 1998).

WHO DEFINED IT ORIGINALLY

Orit Gadiesh and James L. Gilbert of Bain & Company introduced profit-pool mapping in two Harvard Business Review articles in 1998. The method builds on Porter's (1985) value chain and connects to Slywotzky's (1996) contemporaneous work on value migration, with which it is frequently combined.

WHY THIS DEFINITION WAS DEVELOPED

Firms were pursuing revenue share in segments that generated little profit, because revenue was easy to measure and profit by activity was not. Mapping the pool made the misallocation visible, and revealed the recurring pattern in which the largest revenue segment is not the largest profit segment.

FIGURE 1 - PROFIT DOES NOT FOLLOW REVENUE

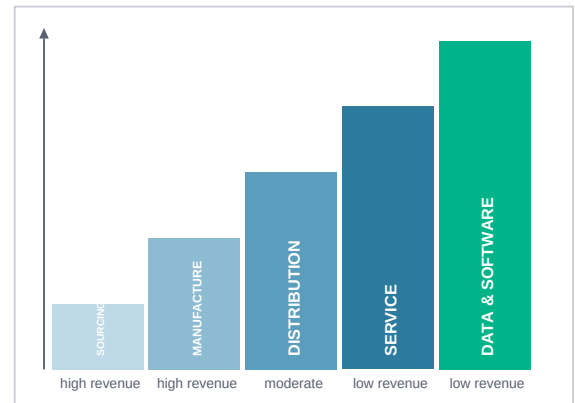


Figure 1. Bar height represents profit; the labels beneath represent revenue share. The characteristic finding is an inverse relationship at one or both ends of the chain, and AI typically shifts profit rightwards toward whichever activity has become the new bottleneck (after Gadiesh and Gilbert, 1998).

HOW IT IS USED

Define the chain's boundaries, estimate revenue and operating margin for each activity, plot margin against revenue share, and identify which activity a new technology is likely to make scarce - because scarcity, not volume, determines where profit settles.

WHEN IT IS USED

In annual strategic review, when an AI capability changes which activity in the chain is scarce, at entry or exit decisions, and when profitability falls while revenue holds.

BY WHOM IT IS USED

Corporate strategy teams, business unit leaders assessing where to compete, investment committees judging vertical moves, and boards examining margin erosion.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The value chain decomposed activities and located cost, but it did not attach profit to each activity or compare across competitors within the chain. Profit-pool mapping supplies that quantification and makes the reallocation dynamic visible over time. In the AI context this is directly consequential: where a model automates a previously scarce human activity, profit does not disappear but moves to whichever adjacent activity has become the new bottleneck.

RELATED DEFINITIONS IN THIS COURSE

[Value Migration - Unit 1](#)
[Competitive Advantage - Unit 3](#)
[Scalable AI Business Model - Unit 3](#)
[Business Model Innovation - Unit 1](#)
[Labour Market Displacement - Unit 5](#)

SOURCES (HARVARD REFERENCING STYLE)

Gadiesh, O. and Gilbert, J.L. (1998) 'Profit pools: a fresh look at strategy', Harvard Business Review, 76(3), pp. 139-147.

Gadiesh, O. and Gilbert, J.L. (1998) 'How to map your industry's profit pool', Harvard Business Review, 76(3), pp. 149-162.

Porter, M.E. (1985) Competitive Advantage: Creating and Sustaining Superior Performance. New York: Free Press.

Slywotzky, A.J. (1996) Value Migration: How to Think Several Moves Ahead of the Competition. Boston, MA: Harvard Business School Press.

AI Portfolio Management

Allocating scarce delivery capacity across AI initiatives by risk and time horizon

GLOSSARY 03 - SHEET 8 / 8

PROFESSIONAL DEFINITION

AI portfolio management is the allocation of an organisation's scarce data, engineering and change-management capacity across a set of AI initiatives that differ in risk, time to value and strategic horizon. It requires that initiatives be balanced across horizons - core improvement, adjacent extension and transformational bets - and that the constraint being allocated be delivery capacity rather than capital, since capital is rarely the binding limit (Baghai, Coley and White, 1999; Nagji and Tuff, 2012).

WHO DEFINED IT ORIGINALLY

The three-horizon structure was set out by Mehrdad Baghai, Stephen Coley and David White of McKinsey & Company in *The Alchemy of Growth* (1999). Bansi Nagji and Geoff Tuff (2012) reformulated it as the innovation ambition matrix in the *Harvard Business Review* and reported the approximate 70-20-10 allocation observed among firms with superior returns.

WHY THIS DEFINITION WAS DEVELOPED

AI initiative portfolios accumulate at the low-risk end, because each individual proposal is easier to approve there, and firms therefore hold a hundred efficiency pilots and no strategic option. A horizon-balanced structure makes the imbalance visible at portfolio level, where it can be corrected, rather than at proposal level, where each decision looks reasonable.

FIGURE 1 - THREE HORIZONS OF AI INVESTMENT



Figure 1. The three horizons run concurrently and are judged by different metrics. Applying horizon-one hurdle rates across the portfolio removes horizons two and three within about two planning cycles, which is the usual reason a firm holds many pilots and no options (after Baghai, Coley and White, 1999).

HOW IT IS USED

Classify each initiative by horizon, compare the resulting distribution against the intended allocation, and treat the gap as the decision; then govern each horizon differently, since a horizon-three bet judged on horizon-one metrics will always be cancelled.

WHEN IT IS USED

At annual planning, when the pilot count rises without production deployments, at quarterly portfolio review, and when a promising initiative has been terminated for failing to show a near-term return.

BY WHOM IT IS USED

Investment committees, transformation and portfolio offices, chief data and AI officers defending long-horizon work, and finance functions setting the metrics by which each horizon is judged.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor tool was the stage-gate process, which assessed each project independently against a common hurdle rate and thereby ensured the systematic elimination of long-horizon options. Three-horizon portfolio management assesses the distribution rather than the project, and applies horizon-specific metrics, so that an initiative with no near-term revenue is judged on learning and option value rather than on a discounted cash flow it cannot produce.

RELATED DEFINITIONS IN THIS COURSE

AI Strategy - Unit 3

AI Roadmap - Unit 4

Proof of Concept and Pilot Purgatory - Unit 4

Benefits Realisation - Unit 4

AI Operating Model - Unit 3

SOURCES (HARVARD REFERENCING STYLE)

Baghai, M., Coley, S. and White, D. (1999) *The Alchemy of Growth: Practical Insights for Building the Enduring Enterprise*. Reading, MA: Perseus Books.

Cooper, R.G. (1990) 'Stage-gate systems: a new tool for managing new products', *Business Horizons*, 33(3), pp. 44-54.

Nagji, B. and Tuff, G. (2012) 'Managing your innovation portfolio', *Harvard Business Review*, 90(5), pp. 66-74.

AI Adoption

GLOSSARY 04 - SHEET 1 / 8

The process by which an approved AI system becomes routine practice in the operating business

PROFESSIONAL DEFINITION

AI adoption is the process by which an approved artificial intelligence system passes from availability into routine use in the operating business, such that the work is genuinely performed differently and the intended benefit is realised. Adoption is distinguished from deployment, which is a technical event, and from acceptance, which is an attitude; only adoption is observable in changed behaviour and changed outcomes (Rogers, 2003; Fichman, 2000).

WHO DEFINED IT ORIGINALLY

The organisational adoption construct derives from Everett M. Rogers's diffusion of innovations (1962; 5th edn 2003) and from Robert G. Fichman's (2000) synthesis of the information-technology diffusion literature, which distinguished organisational assimilation from individual acceptance. The two-stage structure of adoption decision followed by implementation is Zaltman, Duncan and Holbek's (1973).

WHY THIS DEFINITION WAS DEVELOPED

Firms measured deployment - systems in production - and reported it as adoption, which concealed the gap in which most value is lost. A construct anchored in changed practice rather than in system availability was required to make that gap measurable and therefore manageable.

FIGURE 1 - WHERE THE VALUE IS LOST

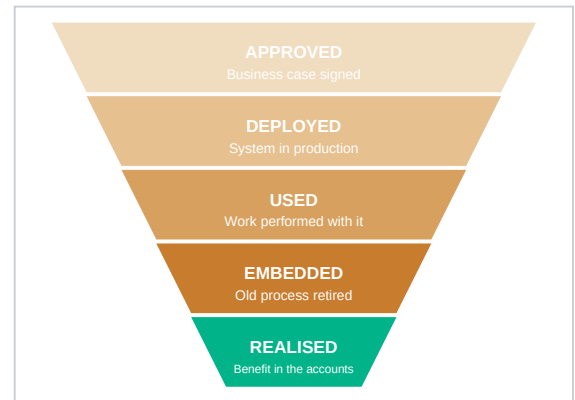


Figure 1. Most AI programmes are measured at the second stage and reported as complete there. The attrition between deployment and embedding is not a technical loss and cannot be recovered by technical means (after Fichman, 2000).

HOW IT IS USED

Measure adoption at the level of the task, using rates of actual use and, more importantly, the operational outcome the system was intended to change; treat any gap between deployment and use as the object of management attention.

WHEN IT IS USED

From the first day of availability rather than at the end of the project, at every stage-gate review, and when benefits promised in the business case have not appeared despite a working system.

BY WHOM IT IS USED

Adoption and change leads, business unit managers whose teams must work differently, benefits owners, and the programme office reporting realisation to the executive.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor measure was go-live: a project was complete when the system entered production, and any subsequent shortfall was treated as a user problem. Framing adoption as a distinct organisational process with its own owner, timeline and budget makes the post-deployment period a managed phase rather than an unmanaged residue, and reallocates resources to the point where value is actually created or lost.

RELATED DEFINITIONS IN THIS COURSE

Change Management - Unit 4

Technology Acceptance Model - Unit 4

Technology Adoption Lifecycle - Unit 1

Benefits Realisation - Unit 4

Organisational Culture - Unit 2

SOURCES (HARVARD REFERENCING STYLE)

Fichman, R.G. (2000) 'The diffusion and assimilation of information technology innovations', in Zmud, R.W. (ed.) Framing the Domains of IT Management. Cincinnati, OH: Pinnaflex, pp. 105-127.

Rogers, E.M. (2003) Diffusion of Innovations. 5th edn. New York: Free Press.

Zaltman, G., Duncan, R. and Holbek, J. (1973) Innovations and Organizations. New York: John Wiley & Sons.

AI Roadmap

A time-phased plan linking AI initiatives to capability, dependency and business outcome

GLOSSARY 04 - SHEET 2 / 8

PROFESSIONAL DEFINITION

An AI roadmap is a time-phased plan that links a sequence of AI initiatives to the capabilities each requires, the dependencies between them, and the business outcomes each is intended to produce. It is a coordination instrument rather than a schedule: its function is to make visible which capability must exist before which initiative can proceed, so that sequencing errors are detected in planning rather than in delivery (Phaal, Farrukh and Probert, 2004).

WHO DEFINED IT ORIGINALLY

Technology roadmapping was developed at Motorola under Robert Galvin in the late 1970s and formalised as the T-Plan method by Robert Phaal, Clare Farrukh and David Probert at the University of Cambridge Institute for Manufacturing, published in Technological Forecasting and Social Change in 2004.

WHY THIS DEFINITION WAS DEVELOPED

Technology plans and business plans were produced by different functions on different cycles and were reconciled only when a delivery date was missed. Roadmapping placed market, product and technology layers on a single time axis so that the dependency between them could be inspected before commitment.

FIGURE 1 - THREE LAYERS ON ONE TIME AXIS



Figure 1. The roadmap is read vertically as well as horizontally. An initiative whose enabling capability arrives after its own start date is a scheduling error visible in planning; discovered in delivery it is an overrun (after Phaal, Farrukh and Probert, 2004).

HOW IT IS USED

Lay out three layers - business outcome, initiative, enabling capability - on one horizontal time axis, and read upward from capability to outcome to test whether each initiative's prerequisites will exist when it starts.

WHEN IT IS USED

At the start of a multi-year AI programme, at annual replanning, when a delivery date slips for reasons of missing data or platform rather than model work, and when a new initiative is proposed mid-cycle.

BY WHOM IT IS USED

Chief data and AI officers, portfolio and programme offices, platform and data engineering leaders supplying the capability layer, and business sponsors owning the outcome layer.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor artefact was the project Gantt chart, which sequences tasks within one initiative and is silent about dependencies across initiatives or about the capability each presupposes. The multi-layer roadmap makes the cross-initiative dependency explicit and forces the capability layer to be stated, which is precisely where AI programmes fail: the model is ready and the governed data feed is not.

RELATED DEFINITIONS IN THIS COURSE

AI Strategy - Unit 3

AI Portfolio Management - Unit 3

MLOps - Unit 4

Benefits Realisation - Unit 4

Digital Maturity - Unit 1

SOURCES (HARVARD REFERENCING STYLE)

Phaal, R., Farrukh, C.J.P. and Probert, D.R. (2004) 'Technology roadmapping: a planning framework for evolution and revolution', Technological Forecasting and Social Change, 71(1-2), pp. 5-26.

Phaal, R., Farrukh, C.J.P. and Probert, D.R. (2001) T-Plan: The Fast Start to Technology Roadmapping. Cambridge: University of Cambridge Institute for Manufacturing.

Willyard, C.H. and McClees, C.W. (1987) 'Motorola's technology roadmap process', Research Management, 30(5), pp. 13-19.

Proof of Concept and Pilot Purgatory

GLOSSARY 04 - SHEET 3 / 8

A bounded feasibility test, and the failure state in which pilots never reach production

PROFESSIONAL DEFINITION

A proof of concept is a small, time-boxed exercise designed to establish whether a proposed approach is technically feasible, and it is complete when that question is answered either way. Pilot purgatory is the organisational failure state in which a portfolio of such exercises accumulates, each judged successful on its own terms, while none reaches production - because the criterion applied was feasibility rather than scalability, and no exit decision was ever required (World Economic Forum and McKinsey, 2018).

WHO DEFINED IT ORIGINALLY

Proof of concept is a general engineering practice without a single originator. The pilot purgatory diagnosis was named by the World Economic Forum and McKinsey & Company in their Global Lighthouse Network research on Fourth Industrial Revolution manufacturing deployments, and has been elaborated in subsequent editions of that programme.

WHY THIS DEFINITION WAS DEVELOPED

The proof of concept is easy to approve, cheap, low-risk and locally rewarded, which produces a portfolio that grows without ever consuming the scarce resource that production deployment requires. Naming the accumulation as a distinct pathology made it a portfolio-level governance failure rather than a series of individually reasonable decisions.

FIGURE 1 - TWO EXITS FROM A PILOT

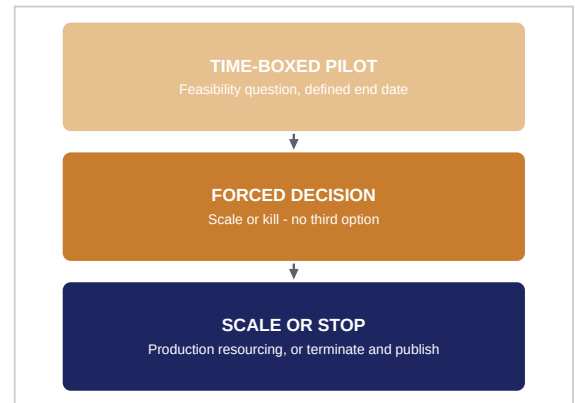


Figure 1. Purgatory is the absence of the middle block. Where no forced decision exists, a pilot neither scales nor stops; it persists at low cost, consumes attention, and is reported as evidence of progress (after World Economic Forum and McKinsey, 2018).

HOW IT IS USED

Define the exit criterion before starting - what result would cause this to proceed to production and what result would cause it to stop - and require an explicit kill or scale decision at the time box's end; a pilot with no defined end state does not terminate.

WHEN IT IS USED

At every pilot approval, at quarterly portfolio review, when pilot count exceeds production deployments, and when the same use case has been piloted more than once.

BY WHOM IT IS USED

Investment committees approving pilots, portfolio offices tracking conversion rates, business sponsors accountable for scaling, and the executives to whom pilot counts are reported as progress.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor governance instrument was the stage-gate, which permits progression but does not compel termination, so a pilot may remain indefinitely between gates. The purgatory framing adds the requirement of a forced decision at the time box and shifts the success criterion from technical feasibility - which almost every AI pilot satisfies - to scalability, which most do not. This changes what a pilot is for.

RELATED DEFINITIONS IN THIS COURSE

[AI Portfolio Management - Unit 3](#)
[AI Adoption - Unit 4](#)
[MLOps - Unit 4](#)
[Benefits Realisation - Unit 4](#)
[AI Roadmap - Unit 4](#)

SOURCES (HARVARD REFERENCING STYLE)

World Economic Forum and McKinsey & Company (2018) The Next Economic Growth Engine: Scaling Fourth Industrial Revolution Technologies in Production. Geneva: World Economic Forum.

World Economic Forum and McKinsey & Company (2019) Fourth Industrial Revolution: Beacons of Technology and Innovation in Manufacturing. Geneva: World Economic Forum.

Cooper, R.G. (1990) 'Stage-gate systems: a new tool for managing new products', Business Horizons, 33(3), pp. 44-54.

Verification note: figures on pilot-to-production conversion rates circulate widely in practitioner literature and vary substantially by source, sector and definition. Cross-check any specific percentage against the primary report and its stated methodology before using it in assessed work.

Change Management

The structured approach to moving individuals and organisations to a future state

GLOSSARY 04 - SHEET 4 / 8

PROFESSIONAL DEFINITION

Change management is the structured approach by which individuals, teams and organisations are moved from a current state to a defined future state so that an intended benefit is realised. Where change leadership addresses direction and motivation, change management addresses the planning, sequencing, resourcing, communication and reinforcement of the transition, and it operates on the individual as the unit of change (Hiatt, 2006; Kotter, 1996).

WHO DEFINED IT ORIGINALLY

The individual-level model most used in practice is ADKAR, developed by Jeffrey M. Hiatt at Prosci from research across more than 700 organisations and published in 2006. The organisational-level antecedents are Lewin (1947) and Kotter (1996); Bridges (1991) supplies the psychological transition account.

WHY THIS DEFINITION WAS DEVELOPED

Organisational change models specified what the enterprise must do but not what must be true of each individual for the change to occur. Hiatt's contribution was to make the individual the unit of analysis and to sequence the five conditions, so that a stalled change could be diagnosed by identifying which condition was absent rather than by repeating the whole programme.

FIGURE 1 - FIVE SEQUENTIAL CONDITIONS

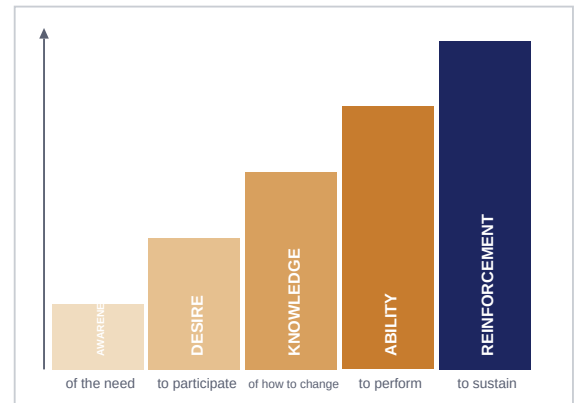


Figure 1. The conditions are sequential, not parallel. Training delivers knowledge and ability; where desire is the missing condition, additional training produces competent non-adopters and no behavioural change (after Hiatt, 2006).

HOW IT IS USED

Diagnose which of the five sequential conditions - awareness, desire, knowledge, ability, reinforcement - is missing for the affected population, and address that one; effort spent on later conditions while an earlier one is absent produces no movement.

WHEN IT IS USED

From the design phase of an AI initiative rather than before go-live, when usage stalls after training, when a change reverts after attention moves elsewhere, and at every significant process redesign.

BY WHOM IT IS USED

Change managers and human resources partners, line managers who deliver reinforcement, project sponsors funding the change effort, and adoption leads diagnosing stalled deployments.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Lewin's three-step model described the organisation's state transitions and offered no diagnostic granularity; a stalled change could only be re-run. ADKAR decomposes the transition into five conditions that are individually observable and strictly sequential, which converts change management from a communications programme into a diagnosis. Its practical value in AI adoption is that it distinguishes cannot from will not - knowledge and ability gaps from desire gaps - which require different remedies.

RELATED DEFINITIONS IN THIS COURSE

Change Leadership - Unit 2

AI Adoption - Unit 4

Technology Acceptance Model - Unit 4

Reskilling and Upskilling - Unit 4

Organisational Culture - Unit 2

SOURCES (HARVARD REFERENCING STYLE)

Bridges, W. (1991) Managing Transitions: Making the Most of Change. Reading, MA: Addison-Wesley.
 Hiatt, J.M. (2006) ADKAR: A Model for Change in Business, Government and Our Community. Loveland, CO: Prosci Learning Center Publications.
 Kotter, J.P. (1996) Leading Change. Boston, MA: Harvard Business School Press.
 Lewin, K. (1947) 'Frontiers in group dynamics', Human Relations, 1(1), pp. 5-41.

MLOps

The engineering discipline for reliably deploying, monitoring and maintaining models in production

GLOSSARY 04 - SHEET 5 / 8

PROFESSIONAL DEFINITION

MLOps is the set of engineering practices for reliably deploying, monitoring, retraining and retiring machine-learning models in production. It extends DevOps by adding the two dependencies that distinguish machine-learning systems from conventional software: the behaviour of the system depends on data as well as on code, and the system degrades over time as the world it was trained on changes, without any change to the code (Sculley et al., 2015; Kreuzberger, Kuhl and Hirschl, 2023).

WHO DEFINED IT ORIGINALLY

The problem was characterised by D. Sculley and colleagues at Google in the 2015 NeurIPS paper on hidden technical debt in machine-learning systems. The term itself emerged from practitioner communities from around 2018; Dominik Kreuzberger, Niklas Kuhl and Sebastian Hirschl (2023) provided the first systematic academic definition and architecture in IEEE Access.

WHY THIS DEFINITION WAS DEVELOPED

Sculley and colleagues demonstrated that the model code is a small fraction of a real machine-learning system, and that the surrounding infrastructure - data collection, feature extraction, serving, monitoring - accumulates a distinctive form of technical debt. A discipline was needed because software engineering practice assumed that a passing test today implies a passing test next quarter, which is untrue of models.

FIGURE 1 - THE CONTINUOUS MODEL LIFECYCLE

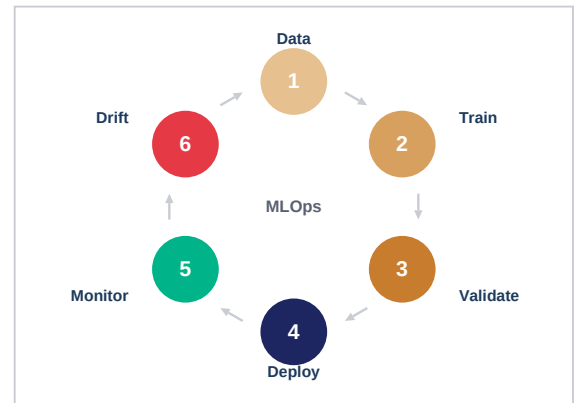


Figure 1. The loop does not close at deployment. Drift detection returns the system to data and retraining without a code change, which is the property that distinguishes MLOps from DevOps and the reason a released model still requires an owner (after Kreuzberger, Kuhl and Hirschl, 2023).

HOW IT IS USED

Version data and models alongside code, automate the retraining and deployment pipeline, monitor for data and concept drift in production, and define the retraining trigger and the rollback path before the first deployment.

WHEN IT IS USED

Before the first model reaches production rather than after the first incident, at every model release, when performance degrades without a code change, and at audit of a deployed high-risk system.

BY WHOM IT IS USED

Machine-learning and platform engineers, data science leads, model risk and validation functions, and the operations teams who carry the system after the project closes.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

DevOps unified development and operations for systems whose behaviour is fully determined by code, and its continuous-integration practice assumes deterministic tests. MLOps adds data and model versioning as first-class artefacts and introduces continuous training triggered by observed drift, which has no DevOps equivalent. Without it, a model that passed validation at release degrades silently, and the first indication is a business outcome rather than an alert.

RELATED DEFINITIONS IN THIS COURSE

Data Strategy - Unit 3

AI Operating Model - Unit 3

Proof of Concept and Pilot Purgatory - Unit 4

AI Governance - Unit 6

Explainability - Unit 6

SOURCES (HARVARD REFERENCING STYLE)

- Kreuzberger, D., Kuhl, N. and Hirschl, S. (2023) 'Machine learning operations (MLOps): overview, definition, and architecture', IEEE Access, 11, pp. 31866-31879.
- Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F. and Dennison, D. (2015) 'Hidden technical debt in machine learning systems', Advances in Neural Information Processing Systems, 28, pp. 2503-2511.
- Treveil, M. and the Dataiku Team (2020) Introducing MLOps. Sebastopol, CA: O'Reilly Media.

Reskilling and Upskilling

Training for a different role, and deepening capability within the current one

GLOSSARY 04 - SHEET 6 / 8

PROFESSIONAL DEFINITION

Upskilling is the development of additional capability within a person's existing role so that the role can be performed to a changed standard. Reskilling is the development of the capability required for a materially different role, undertaken because the original role has been substantially altered or eliminated. The distinction matters because the two have different costs, durations, success rates and organisational obligations (World Economic Forum, 2023; Illanes et al., 2018).

WHO DEFINED IT ORIGINALLY

The two-term distinction was consolidated in policy and management use by the World Economic Forum's Future of Jobs report series from 2016 onward, and in management practice by Pablo Illanes, Susan Lund, Mona Mourshed, Scott Rutherford and Magnus Tyreman of McKinsey & Company (2018). The economic foundation is Becker's (1964) distinction between general and firm-specific human capital.

WHY THIS DEFINITION WAS DEVELOPED

Firms facing automation described a single training requirement and budgeted for it at upskilling cost while the actual need was reskilling, which is substantially more expensive, takes longer and fails more often. Separating the terms allowed workforce plans to be costed against the transition actually required.

FIGURE 1 - TWO DIFFERENT TRANSITIONS

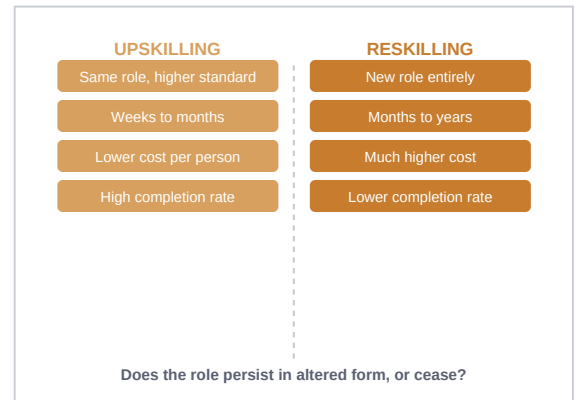


Figure 1. The two transitions are routinely budgeted at the same unit cost. Where a role ceases rather than changes, an upskilling budget funds a reskilling requirement and the shortfall appears as attrition rather than as a training variance (after Illanes et al., 2018).

HOW IT IS USED

Determine for each affected role whether the role persists in altered form or ceases; upskill in the first case, reskill in the second, and cost and time each separately rather than as a single training line.

WHEN IT IS USED

During AI workforce planning, before rather than after automation is deployed, at annual capability review, and when a role's task composition changes by more than a modest fraction.

BY WHOM IT IS USED

Chief human resources officers, workforce planning teams, line managers assessing role composition, works councils and employee representatives, and the individuals whose roles are changing.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor category was training, an undifferentiated budget line whose unit cost was assumed constant across purposes. Distinguishing the two makes the workforce plan a portfolio with different transition costs per person, and connects to Becker's insight about who pays: firm-specific capability is rationally funded by the employer, general capability is not, which explains the persistent under-investment in reskilling precisely where it is most needed.

RELATED DEFINITIONS IN THIS COURSE

AI Literacy - Unit 2

Change Management - Unit 4

Labour Market Displacement - Unit 5

Skill-Biased Technological Change - Unit 5

AI Adoption - Unit 4

SOURCES (HARVARD REFERENCING STYLE)

Becker, G.S. (1964) Human Capital: A Theoretical and Empirical Analysis. New York: Columbia University Press.

Illanes, P., Lund, S., Mourshed, M., Rutherford, S. and Tyreman, M. (2018) Retraining and Reskilling Workers in the Age of Automation. New York: McKinsey Global Institute.

World Economic Forum (2023) Future of Jobs Report 2023. Geneva: World Economic Forum.

Technology Acceptance Model

GLOSSARY 04 - SHEET 7 / 8

Perceived usefulness and perceived ease of use predict intention to use a system

PROFESSIONAL DEFINITION

The Technology Acceptance Model holds that an individual's intention to use an information system is determined principally by two beliefs: perceived usefulness, the degree to which the person believes the system will enhance their job performance, and perceived ease of use, the degree to which they believe using it will be free of effort. Perceived ease of use also influences perceived usefulness, and usefulness is the stronger predictor of the two (Davis, 1989).

WHO DEFINED IT ORIGINALLY

Fred D. Davis developed the model in his 1985 MIT Sloan doctoral dissertation and published it in MIS Quarterly in 1989. It adapts Fishbein and Ajzen's (1975) theory of reasoned action to the technology context. Venkatesh and Davis (2000) extended it as TAM2, and Venkatesh, Morris, Davis and Davis (2003) consolidated eight competing models into UTAUT.

WHY THIS DEFINITION WAS DEVELOPED

Organisations were investing in systems that users declined to use, and the available explanations were post hoc. Davis sought a model that would predict acceptance from beliefs measurable before deployment, so that design and communication could be corrected while correction was still cheap.

HOW IT IS USED

Measure the two beliefs among the intended user population during design or pilot; a usefulness deficit requires a change to the system's fit to the job, an ease-of-use deficit requires a change to interface or training, and the two are not interchangeable.

WHEN IT IS USED

During design of an AI decision-support tool, at pilot before scaling, when usage is below plan after go-live, and when comparing two candidate vendor products for the same task.

BY WHOM IT IS USED

Product owners and user-experience designers, adoption and change leads, procurement teams evaluating vendor systems, and business sponsors deciding whether to scale a pilot.

FIGURE 1 - TWO BELIEFS, ONE BEHAVIOUR

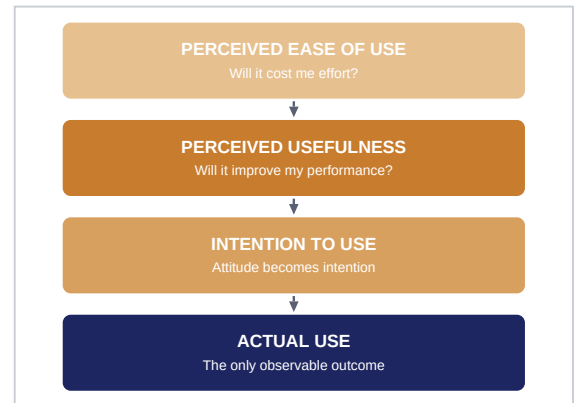


Figure 1. Ease of use acts on behaviour partly through usefulness, which is the stronger predictor. A system that is effortless and not useful is not adopted; the reverse is frequently adopted despite complaint (after Davis, 1989).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Its predecessor, the theory of reasoned action, was a general model of behaviour requiring a fresh belief-elicitation study for each context, which made it impractical for system evaluation. Davis fixed the two belief constructs in advance and supplied validated scales, so acceptance could be measured comparably across systems and organisations. UTAUT later added social influence and facilitating conditions, which matter particularly where use is mandated rather than voluntary - the usual case in enterprise AI.

RELATED DEFINITIONS IN THIS COURSE

[AI Adoption - Unit 4](#)
[Algorithm Aversion - Unit 2](#)
[Change Management - Unit 4](#)
[Technology Adoption Lifecycle - Unit 1](#)
[AI Literacy - Unit 2](#)

SOURCES (HARVARD REFERENCING STYLE)

Davis, F.D. (1989) 'Perceived usefulness, perceived ease of use, and user acceptance of information technology', MIS Quarterly, 13(3), pp. 319-340.
 Fishbein, M. and Ajzen, I. (1975) Belief, Attitude, Intention and Behavior. Reading, MA: Addison-Wesley.
 Venkatesh, V. and Davis, F.D. (2000) 'A theoretical extension of the technology acceptance model', Management Science, 46(2), pp. 186-204.
 Venkatesh, V., Morris, M.G., Davis, G.B. and Davis, F.D. (2003) 'User acceptance of information technology: toward a unified view', MIS Quarterly, 27(3), pp. 425-478.

Benefits Realisation

GLOSSARY 04 - SHEET 8 / 8

The management process ensuring that promised benefits are defined, owned and delivered

PROFESSIONAL DEFINITION

Benefits realisation management is the process of identifying, defining, planning, tracking and reviewing the benefits attributed to an investment, so that they are actually delivered rather than merely forecast. Each benefit must have a named owner outside the delivery team, a baseline measurement taken before the change, and a defined measurement point after it; benefits without all three are unverifiable claims (Ward and Daniel, 2012; Bradley, 2010).

WHO DEFINED IT ORIGINALLY

The benefits dependency network and the associated management process were developed by John Ward and Elizabeth Daniel at Cranfield School of Management, published as Benefits Management (2006; 2nd edn 2012). Gerald Bradley (2010) supplied a parallel practitioner treatment. The method is embedded in the UK Office of Government Commerce's Managing Successful Programmes.

WHY THIS DEFINITION WAS DEVELOPED

Business cases were produced to obtain approval and then archived, with no mechanism connecting forecast benefit to any subsequent measurement. Ward and Daniel argued that benefits arise from business change rather than from technology, and that the change producing each benefit must be identified and owned if the benefit is to occur at all.

HOW IT IS USED

Build the dependency network backwards from the intended benefit through the business changes required to produce it to the technology enabling those changes, and assign each benefit an owner in the business who is measured on it.

WHEN IT IS USED

At business-case approval, at every stage-gate, at a defined post-implementation review, and at the point where an AI system is declared complete by the delivery team.

BY WHOM IT IS USED

Benefits owners in the operating business, programme and portfolio offices, finance functions validating claimed benefits, and executive sponsors accountable for the investment.

FIGURE 1 - THE BENEFITS DEPENDENCY NETWORK

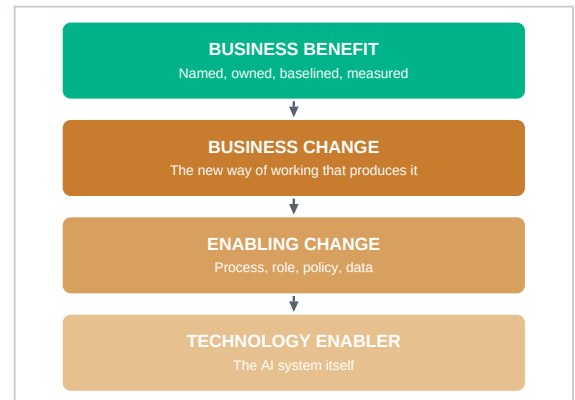


Figure 1. The network is built from the top downward and delivered from the bottom upward. Reading it downward from the benefit exposes business changes that nobody has agreed to make, which is where forecast benefits are lost (after Ward and Daniel, 2012).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor instrument was the business case, prepared once for approval, with no owner after the funding decision and no measurement discipline. Benefits realisation makes the benefit an object with a lifecycle, an owner and a baseline, and its dependency network reverses the usual logic: it starts from the benefit and derives the technology, rather than starting from the technology and asserting benefits. For AI investments, where benefit claims are frequently generic, this is the difference between a forecast and a commitment.

RELATED DEFINITIONS IN THIS COURSE

[AI Adoption - Unit 4](#)
[AI Portfolio Management - Unit 3](#)
[Proof of Concept and Pilot Purgatory - Unit 4](#)
[AI Roadmap - Unit 4](#)
[Externality - Unit 5](#)

SOURCES (HARVARD REFERENCING STYLE)

Bradley, G. (2010) Benefit Realisation Management: A Practical Guide to Achieving Benefits Through Change. 2nd edn. Farnham: Gower.

Ward, J. and Daniel, E. (2012) Benefits Management: How to Increase the Business Value of Your IT Projects. 2nd edn. Chichester: John Wiley & Sons.

Office of Government Commerce (2011) Managing Successful Programmes. 4th edn. London: The Stationery Office.

Productivity Paradox

Measured productivity gains lag the deployment of a transformative technology

GLOSSARY 05 - SHEET 1 / 8

PROFESSIONAL DEFINITION

The productivity paradox is the observed discrepancy between rapid deployment of a powerful information technology and the absence of a corresponding rise in measured aggregate productivity. The modern explanation is that general purpose technologies require large, slow and unmeasured investments in complementary intangible assets - process redesign, skills, organisational capital - which depress measured productivity while they are being made and raise it only afterwards (Solow, 1987; Brynjolfsson, Rock and Syverson, 2021).

WHO DEFINED IT ORIGINALLY

Robert M. Solow's 1987 New York Times Book Review remark that computers were visible everywhere except in the productivity statistics named the phenomenon. Erik Brynjolfsson and Lorin Hitt (1996, 2003) at MIT established the firm-level resolution; Brynjolfsson, Daniel Rock and Chad Syverson (2021) formalised the J-curve account in the American Economic Journal: Macroeconomics.

WHY THIS DEFINITION WAS DEVELOPED

The paradox had been used to argue either that the technology was overrated or that measurement was defective. The J-curve account supplies a third explanation with a testable implication: firms making the complementary intangible investment record lower measured productivity during the investment period and higher productivity afterwards, so the aggregate lag is a composition effect rather than a measurement error.

FIGURE 1 - THE PRODUCTIVITY J-CURVE

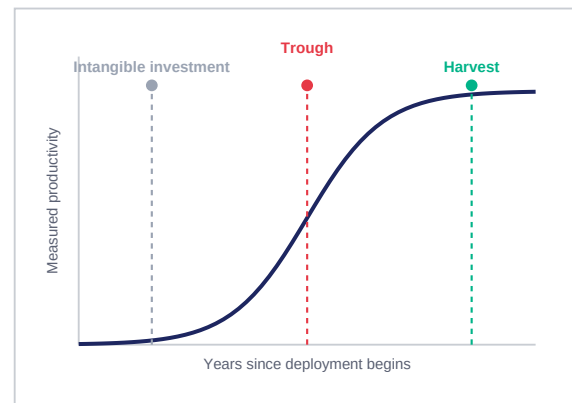


Figure 1. Measured productivity falls before it rises, because the complementary intangible investment is expensed while its output is not yet capitalised. Appraisal horizons shorter than the full curve observe only the decline (after Brynjolfsson, Rock and Syverson, 2021).

HOW IT IS USED

When appraising an AI investment, budget explicitly for the complementary intangible investment and set the measurement window beyond the trough; a two-year appraisal horizon will observe only the negative portion of the curve.

WHEN IT IS USED

At business-case appraisal for large AI programmes, in board discussion of why the technology has not yet appeared in results, at national or sectoral policy analysis, and when investment is threatened by disappointing interim returns.

BY WHOM IT IS USED

Chief financial officers setting appraisal horizons, economists and policy analysts, board members interpreting interim performance, and transformation directors defending multi-year programmes.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor account was mismeasurement: output was said to be rising in ways the national accounts did not capture. The J-curve model retains mismeasurement but relocates it to the asset side - the intangible investment is expensed rather than capitalised - which generates a specific, falsifiable time path rather than a general caution. It also yields a managerial prescription that the mismeasurement account did not: budget the complement, extend the window.

RELATED DEFINITIONS IN THIS COURSE

General Purpose Technology - Unit 5

Benefits Realisation - Unit 4

Digital Maturity - Unit 1

Skill-Biased Technological Change - Unit 5

AI Adoption - Unit 4

SOURCES (HARVARD REFERENCING STYLE)

- Brynjolfsson, E. and Hitt, L.M. (1996) 'Paradox lost? Firm-level evidence on the returns to information systems spending', *Management Science*, 42(4), pp. 541-558.
- Brynjolfsson, E., Rock, D. and Syverson, C. (2021) 'The productivity J-curve: how intangibles complement general purpose technologies', *American Economic Journal: Macroeconomics*, 13(1), pp. 333-372.
- Solow, R.M. (1987) 'We'd better watch out', *New York Times Book Review*, 12 July, p. 36.
- Syverson, C. (2017) 'Challenges to mismeasurement explanations for the US productivity slowdown', *Journal of Economic Perspectives*, 31(2), pp. 165-186.

General Purpose Technology

A pervasive technology that improves over time and spawns complementary innovation

GLOSSARY 05 - SHEET 2 / 8

PROFESSIONAL DEFINITION

A general purpose technology is a technology characterised by three properties: pervasiveness across many sectors, continuing improvement over time that lowers the cost of use, and the capacity to spawn complementary innovation in the sectors that adopt it. Steam power, electricity and the semiconductor are the canonical cases; the economic significance of such technologies lies less in their direct output than in the innovation they enable elsewhere (Bresnahan and Trajtenberg, 1995).

WHO DEFINED IT ORIGINALLY

Timothy F. Bresnahan and Manuel Trajtenberg defined the category in the Journal of Econometrics in 1995. Elhanan Helpman (1998) assembled the growth-theoretic treatment; Paul A. David's (1990) study of the dynamo established the long delay between availability and productivity effect. Application to AI is developed by Agrawal, Gans and Goldfarb (2018) and Brynjolfsson, Rock and Syverson (2021).

WHY THIS DEFINITION WAS DEVELOPED

Growth accounting treated technologies as separable inputs, which could not explain why a small number of innovations produced economy-wide transformation while most did not. The general purpose category identifies the structural properties that generate innovational complementarities, and thereby explains the characteristic delay: the enabling technology must be widely deployed before the complementary innovations that carry the productivity effect become possible.

FIGURE 1 - THREE DEFINING PROPERTIES

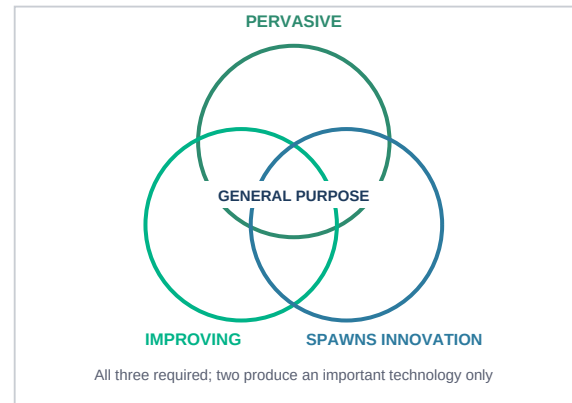


Figure 1. Many technologies satisfy two of the three properties. The third - the capacity to raise the return to innovation in adopting sectors - is what produces economy-wide effect, and it is also the source of the characteristic multi-decade delay (after Bresnahan and Trajtenberg, 1995).

HOW IT IS USED

Test the three properties against the technology in question, then plan on the historical delay pattern rather than on vendor timelines; the value accrues to complementary innovation in the adopting sector, not to the enabling technology itself.

WHEN IT IS USED

In long-horizon strategic planning, in policy analysis of AI's economic effects, when assessing whether an AI investment should be treated as infrastructure or as an application, and in board discussion of pace.

BY WHOM IT IS USED

Corporate strategists on long horizons, economists and policy analysts, investors assessing structural rather than cyclical change, and executives calibrating expectations about timing.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Its predecessor in the growth literature was Schumpeter's account of innovation clusters, which described bunching in time without specifying the mechanism. Bresnahan and Trajtenberg supply the mechanism as innovational complementarity: the enabling technology raises the return to innovation in the adopting sector, which raises the return to further improvement in the enabling technology. David's dynamo evidence then converts the framework into a timing prior - electrification took some four decades to appear in productivity statistics.

RELATED DEFINITIONS IN THIS COURSE

Artificial Intelligence - Unit 1

Productivity Paradox - Unit 5

Disruptive Innovation - Unit 1

Skill-Biased Technological Change - Unit 5

Digital Divide - Unit 5

SOURCES (HARVARD REFERENCING STYLE)

Agrawal, A., Gans, J. and Goldfarb, A. (2018) Prediction Machines: The Simple Economics of Artificial Intelligence. Boston, MA: Harvard Business Review Press.

Bresnahan, T.F. and Trajtenberg, M. (1995) 'General purpose technologies: engines of growth?', Journal of Econometrics, 65(1), pp. 83-108.

David, P.A. (1990) 'The dynamo and the computer: an historical perspective on the modern productivity paradox', American Economic Review, 80(2), pp. 355-361.

Helpman, E. (ed.) (1998) General Purpose Technologies and Economic Growth. Cambridge, MA: MIT Press.

Skill-Biased Technological Change

GLOSSARY 05 - SHEET 3 / 8

Technical change that raises demand for some skills and lowers it for others

PROFESSIONAL DEFINITION

Skill-biased technological change is technical change that shifts relative demand across skill groups, raising the productivity and therefore the wages of favoured groups relative to others. The refined task-based account holds that the bias operates on tasks rather than on skill levels: technologies substitute for routine tasks - whether manual or cognitive - and complement non-routine tasks, which produces polarisation rather than a simple skill gradient (Autor, Levy and Murnane, 2003; Acemoglu and Autor, 2011).

WHO DEFINED IT ORIGINALLY

The canonical skill-biased account is Katz and Murphy's (1992). David H. Autor, Frank Levy and Richard J. Murnane reformulated it as the routine-biased task model in the Quarterly Journal of Economics in 2003; Daron Acemoglu and Autor (2011) developed the general task framework, and Acemoglu and Restrepo (2019) added the displacement-reinstatement distinction.

WHY THIS DEFINITION WAS DEVELOPED

The simple skill-based account predicted monotonically rising returns to education, but observed wage distributions were polarising: employment and wages grew at both ends and hollowed in the middle. Relocating the bias from skill to task explained this, because routine tasks are concentrated in the middle of the wage distribution rather than at its bottom.

FIGURE 1 - SUBSTITUTION AND COMPLEMENTARITY BY TASK

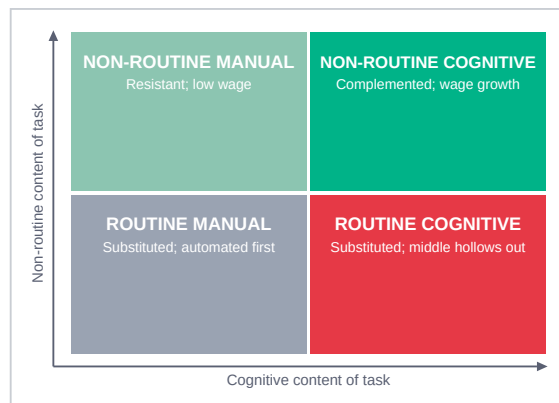


Figure 1. Substitution follows routineness, not skill level. Because routine cognitive tasks sit in the middle of the wage distribution, the effect is polarisation - and generative systems have extended substitution into parts of the upper-right quadrant previously assumed safe (after Autor, Levy and Murnane, 2003).

HOW IT IS USED

Decompose roles into tasks and classify each as routine or non-routine before estimating exposure; the task composition of a role, not its wage level or educational requirement, predicts which parts of it a system will substitute.

WHEN IT IS USED

In workforce planning ahead of AI deployment, in labour-market and policy analysis, in reskilling design, and in public debate about which occupations are exposed.

BY WHOM IT IS USED

Labour economists and policy analysts, chief human resources officers planning workforce transition, works councils, and educators designing curricula.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor model treated skill as a single ordered dimension and could not accommodate polarisation without ad hoc additions. The task framework replaces skill with task as the unit that technology substitutes or complements, which generates polarisation as a prediction rather than an anomaly. Acemoglu and Restrepo's later distinction between displacement and reinstatement effects adds the crucial policy point: whether technology reduces labour demand depends on whether new tasks are created, which is a design and investment choice rather than a technological inevitability.

RELATED DEFINITIONS IN THIS COURSE

Labour Market Displacement - Unit 5

Reskilling and Upskilling - Unit 4

Productivity Paradox - Unit 5

Digital Divide - Unit 5

General Purpose Technology - Unit 5

SOURCES (HARVARD REFERENCING STYLE)

- Acemoglu, D. and Autor, D. (2011) 'Skills, tasks and technologies: implications for employment and earnings', in Ashenfelter, O. and Card, D. (eds) Handbook of Labor Economics, Vol. 4B. Amsterdam: Elsevier, pp. 1043-1171.
- Acemoglu, D. and Restrepo, P. (2019) 'Automation and new tasks: how technology displaces and reinstates labor', Journal of Economic Perspectives, 33(2), pp. 3-30.
- Autor, D.H., Levy, F. and Murnane, R.J. (2003) 'The skill content of recent technological change: an empirical exploration', Quarterly Journal of Economics, 118(4), pp. 1279-1333.
- Katz, L.F. and Murphy, K.M. (1992) 'Changes in relative wages, 1963-1987: supply and demand factors', Quarterly Journal of Economics, 107(1), pp. 35-78.

Labour Market Displacement

GLOSSARY 05 - SHEET 4 / 8

Reduction in labour demand for tasks a technology performs, net of reinstatement

PROFESSIONAL DEFINITION

Labour market displacement is the reduction in demand for labour in tasks that a technology has come to perform. Its net employment effect depends on the balance between the displacement effect, which lowers labour demand at given output, and countervailing effects - the productivity effect, capital accumulation, and above all the reinstatement effect, in which new tasks are created in which labour holds comparative advantage (Acemoglu and Restrepo, 2019).

WHO DEFINED IT ORIGINALLY

The displacement-reinstatement decomposition was developed by Daron Acemoglu of MIT and Pascual Restrepo of Boston University in a sequence of papers from 2018, summarised in the Journal of Economic Perspectives in 2019, with empirical estimates for industrial robots in Acemoglu and Restrepo (2020). Frey and Osborne's (2017) occupational exposure estimates represent the earlier occupation-level tradition.

WHY THIS DEFINITION WAS DEVELOPED

Public debate ran between the claim that automation destroys employment and the claim that it has never done so historically. The decomposition replaces both with an accounting identity in which the net outcome depends on measurable components, and identifies reinstatement as the term that has varied across historical episodes and that is influenced by policy and investment choices.

HOW IT IS USED

Estimate displacement at task level rather than occupation level, then ask separately what new tasks the technology creates and whether the firm or the economy is investing in them; a displacement estimate without a reinstatement estimate is half an answer.

WHEN IT IS USED

In workforce transition planning, in social and economic impact assessment of an AI deployment, in policy analysis, and in works council consultation.

BY WHOM IT IS USED

Policy analysts and labour economists, workforce planners and human resources leaders, employee representatives, and executives accountable for the social impact of deployment decisions.

FIGURE 1 - DISPLACEMENT AGAINST REINSTATEMENT

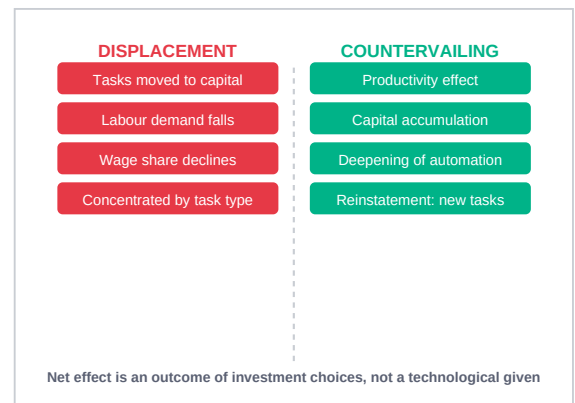


Figure 1. The net employment effect is the balance of the two columns. Historically the right-hand column has dominated, but reinstatement is the term that has varied across episodes and it responds to the direction of investment (after Acemoglu and Restrepo, 2019).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Frey and Osborne's occupation-level probabilities treated whole occupations as automatable and produced headline figures that were widely misread as job-loss forecasts. The task-level decomposition disaggregates the occupation, which is more accurate because most occupations comprise both substitutable and complemented tasks, and it makes reinstatement explicit. The resulting managerial and policy point is that the net effect is partly chosen: technologies can be directed toward automating existing tasks or toward creating new ones.

RELATED DEFINITIONS IN THIS COURSE

Skill-Biased Technological Change - Unit 5

Reskilling and Upskilling - Unit 4

Profit Pool - Unit 3

Externality - Unit 5

Digital Divide - Unit 5

SOURCES (HARVARD REFERENCING STYLE)

- Acemoglu, D. and Restrepo, P. (2019) 'Automation and new tasks: how technology displaces and reinstates labor', Journal of Economic Perspectives, 33(2), pp. 3-30.
- Acemoglu, D. and Restrepo, P. (2020) 'Robots and jobs: evidence from US labor markets', Journal of Political Economy, 128(6), pp. 2188-2244.
- Frey, C.B. and Osborne, M.A. (2017) 'The future of employment: how susceptible are jobs to computerisation?', Technological Forecasting and Social Change, 114, pp. 254-280.

Externality

A cost or benefit borne by a third party who is not party to the transaction

GLOSSARY 05 - SHEET 5 / 8

PROFESSIONAL DEFINITION

An externality is a cost or benefit arising from an economic activity that falls on a party who neither chose to incur it nor is compensated for it, and which therefore does not appear in the price of the activity. Because the decision-maker does not bear the full cost, activity generating negative externalities is over-supplied relative to the socially efficient level; correction requires that the external cost be internalised through regulation, taxation or assignment of rights (Pigou, 1920; Coase, 1960).

WHO DEFINED IT ORIGINALLY

Arthur C. Pigou formalised the concept and the corrective tax in *The Economics of Welfare* (1920) at the University of Cambridge. Ronald H. Coase (1960) established that with well-defined property rights and low transaction costs the parties may bargain to an efficient outcome without a tax, for which he received the 1991 Nobel Memorial Prize.

WHY THIS DEFINITION WAS DEVELOPED

Classical price theory assumed that prices carried all relevant costs, which was contradicted by industrial pollution. Pigou needed a mechanism reconciling private optimisation with social efficiency. Coase's later contribution reframed the question from who is at fault to which allocation of rights minimises total cost, which changed the design of environmental and technology regulation.

FIGURE 1 - PRIVATE COST AGAINST SOCIAL COST



Figure 1. The gap between the two is the externality. Because the deploying firm optimises against private cost, activity generating negative externalities is over-supplied until a rule, a tax or an assignment of rights internalises the difference (after Pigou, 1920; Coase, 1960).

HOW IT IS USED

Identify who bears the costs of an AI deployment other than the deploying firm and its customers - energy and water consumption, labour transition costs, effects on those subject to automated decisions - and state them in the appraisal even where they are not priced.

WHEN IT IS USED

In AI impact assessment, in environmental appraisal of computing infrastructure, in regulatory analysis, and in any business case whose benefits accrue internally while some costs do not.

BY WHOM IT IS USED

Policy analysts and regulators, sustainability and environmental social and governance functions, boards assessing societal impact, and economists appraising public investment.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor position had no separate category for uncompensated third-party effects, which meant they entered analysis only as moral objection. Pigou made the effect a quantity to be measured and priced; Coase then showed that the tax is one of several instruments and not always the efficient one. For AI the framework is directly applicable: energy consumption, labour displacement and error costs borne by decision subjects are all external to the deployment decision unless a rule makes them internal.

RELATED DEFINITIONS IN THIS COURSE

[Green AI - Unit 5](#)
[Labour Market Displacement - Unit 5](#)
[Life Cycle Assessment - Unit 5](#)
[Benefits Realisation - Unit 4](#)
[AI Ethics - Unit 6](#)

SOURCES (HARVARD REFERENCING STYLE)

Coase, R.H. (1960) 'The problem of social cost', *Journal of Law and Economics*, 3, pp. 1-44.

Pigou, A.C. (1920) *The Economics of Welfare*. London: Macmillan.

Stern, N. (2007) *The Economics of Climate Change: The Stern Review*. Cambridge: Cambridge University Press.

Green AI

Treating computational and environmental cost as a reported outcome alongside accuracy

GLOSSARY 05 - SHEET 6 / 8

PROFESSIONAL DEFINITION

Green AI is the practice of treating the computational and environmental cost of machine-learning research and deployment as a first-class evaluation criterion alongside accuracy, and of reporting it. It is contrasted with Red AI, in which accuracy improvements are pursued through increased computation without that cost being reported. The proposed reporting measure is total floating-point operations required to produce a result, which is hardware-independent and comparable across studies (Schwartz et al., 2020).

WHO DEFINED IT ORIGINALLY

Roy Schwartz, Jesse Dodge, Noah A. Smith and Oren Etzioni of the Allen Institute for AI proposed the distinction in a 2019 preprint published in Communications of the ACM in 2020. Emma Strubell, Ananya Ganesh and Andrew McCallum (2019) provided the influential energy estimates for training; Patterson et al. (2021) supplied a more precise accounting of carbon cost.

WHY THIS DEFINITION WAS DEVELOPED

Reported progress was being achieved substantially by increasing computation, with the cost invisible in publication and inaccessible to smaller institutions. Making computational cost a reported outcome was intended to address both the environmental consequence and the concentration of research capability in organisations able to afford the compute.

HOW IT IS USED

Report computational cost alongside performance for every model, and evaluate improvements against cost as well as against accuracy; where the accuracy gain per unit of computation is negligible, the smaller model is the correct deployment choice.

WHEN IT IS USED

At model selection, at procurement of AI services, in sustainability reporting, and when a marginal accuracy improvement requires a disproportionate increase in computation.

BY WHOM IT IS USED

Machine-learning practitioners and research leads, sustainability and reporting functions, procurement teams comparing vendor offerings, and boards with climate commitments.

FIGURE 1 - ACCURACY AGAINST COMPUTATIONAL COST

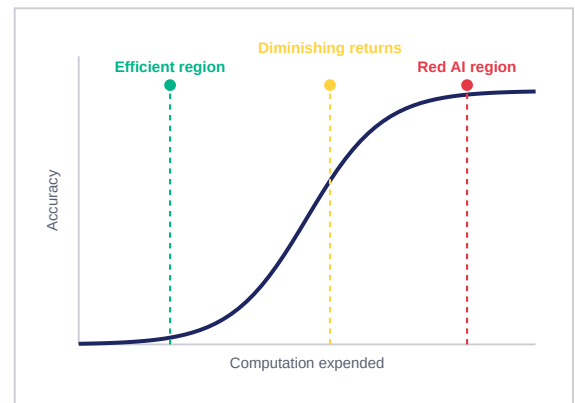


Figure 1. Accuracy rises with computation at a sharply decreasing rate. Reporting only the vertical axis makes movement into the right-hand region look like progress; reporting both axes makes it a cost decision (after Schwartz et al., 2020).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor evaluation norm reported accuracy alone, which made increased computation an invisible and therefore costless input. Green AI adds efficiency to the reported result and proposes a hardware-independent unit, which permits comparison across studies and across time in a way that energy or monetary cost does not. Its limitation is that reporting remains voluntary in most venues, so the distinction currently operates as a norm rather than a requirement.

RELATED DEFINITIONS IN THIS COURSE

Life Cycle Assessment - Unit 5

Externality - Unit 5

MLOps - Unit 4

AI Governance - Unit 6

Productivity Paradox - Unit 5

SOURCES (HARVARD REFERENCING STYLE)

- Patterson, D., Gonzalez, J., Le, Q., Liang, C., Munguia, L.-M., Rothchild, D., So, D., Texier, M. and Dean, J. (2021) Carbon Emissions and Large Neural Network Training. arXiv:2104.10350.
- Schwartz, R., Dodge, J., Smith, N.A. and Etzioni, O. (2020) 'Green AI', Communications of the ACM, 63(12), pp. 54-63.
- Strubell, E., Ganesh, A. and McCallum, A. (2019) 'Energy and policy considerations for deep learning in NLP', in Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics. Stroudsburg, PA: ACL, pp. 3645-3650.

Verification note: published energy and carbon estimates for model training vary by more than an order of magnitude depending on hardware, data-centre efficiency, grid carbon intensity and whether inference is included. Cite the primary study and its stated boundary conditions rather than a secondary summary figure.

Life Cycle Assessment

Standardised accounting of environmental impact across a product's whole life

GLOSSARY 05 - SHEET 7 / 8

PROFESSIONAL DEFINITION

Life cycle assessment is the compilation and evaluation of the inputs, outputs and potential environmental impacts of a product system throughout its life cycle, from raw material extraction through production, use and end-of-life. It is standardised in four phases - goal and scope definition, inventory analysis, impact assessment, and interpretation - by ISO 14040 and ISO 14044, and its results are only comparable where the system boundary and functional unit are identically defined (ISO, 2006).

WHO DEFINED IT ORIGINALLY

The method emerged from packaging studies conducted for the Coca-Cola Company by the Midwest Research Institute in 1969 and was developed through the Society of Environmental Toxicology and Chemistry from the 1990s. It was standardised by the International Organization for Standardization as ISO 14040 and ISO 14044, first published in 1997 and revised in 2006.

WHY THIS DEFINITION WAS DEVELOPED

Environmental claims were being made on the basis of a single life-cycle stage, which permitted burden-shifting - a product could be made cleaner in use by being made dirtier in manufacture, and reported as an improvement. A standardised whole-life boundary made such shifting visible and made competing claims comparable.

FIGURE 1 - THE FOUR PHASES OF AN ASSESSMENT

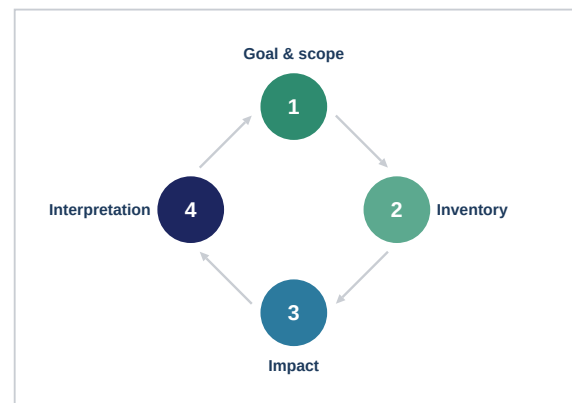


Figure 1. Interpretation feeds back into scope, which is why the method is iterative rather than linear. For AI systems the first phase is decisive: whether hardware manufacture and inference at scale sit inside the boundary determines the result more than the inventory does (after ISO, 2006).

HOW IT IS USED

Define the functional unit and system boundary before collecting any data, since these determine the result more than the inventory does; for AI systems this means deciding explicitly whether hardware manufacture, training and inference are inside the boundary.

WHEN IT IS USED

In sustainability reporting, in procurement of AI infrastructure, when comparing an AI-enabled process against the process it replaces, and when an environmental claim is made about a digital service.

BY WHOM IT IS USED

Sustainability and reporting functions, procurement and supply chain teams, environmental consultants conducting the assessment, and auditors verifying disclosed figures.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor practice was single-issue or single-stage assessment - energy in use, or emissions at the factory gate - which invited burden-shifting between stages and between impact categories. The ISO structure requires a declared functional unit and boundary and mandates the interpretation phase, in which sensitivity to those choices must be examined. For AI this is the decisive discipline, since assessments that exclude hardware manufacture or inference at scale can differ from complete ones by a large factor.

RELATED DEFINITIONS IN THIS COURSE

Green AI - Unit 5

Externality - Unit 5

AI Governance - Unit 6

Benefits Realisation - Unit 4

SOURCES (HARVARD REFERENCING STYLE)

International Organization for Standardization (2006) ISO 14040:2006 Environmental Management - Life Cycle Assessment - Principles and Framework. Geneva: ISO.

International Organization for Standardization (2006) ISO 14044:2006 Environmental Management - Life Cycle Assessment - Requirements and Guidelines. Geneva: ISO.

Hellweg, S. and Mila i Canals, L. (2014) 'Emerging approaches, challenges and opportunities in life cycle assessment', Science, 344(6188), pp. 1109-1113.

Digital Divide

Systematic inequality in access to, use of, and benefit from digital technology

GLOSSARY 05 - SHEET 8 / 8

PROFESSIONAL DEFINITION

The digital divide is the systematic inequality between individuals, groups, regions or states in access to digital technology, in the skills and autonomy to use it effectively, and in the outcomes obtained from that use. The literature distinguishes three levels: the first-level divide in physical access, the second-level divide in skills and usage, and the third-level divide in the tangible benefits realised (van Dijk, 2005; Scheerder, van Deursen and van Dijk, 2017).

WHO DEFINED IT ORIGINALLY

The term entered policy use through the United States National Telecommunications and Information Administration's Falling Through the Net reports from 1995. Jan A.G.M. van Dijk (2005) at the University of Twente developed the resources-and-appropriation theory; Paul DiMaggio and Eszter Hargittai (2001) introduced the second-level divide, and Alexander van Deursen and colleagues the third.

WHY THIS DEFINITION WAS DEVELOPED

Policy assumed that inequality would resolve as connectivity spread, which access statistics eventually confirmed while outcome inequality persisted or widened. Distinguishing three levels explained why: closing the access gap shifts the inequality to skills, and closing the skills gap shifts it to the benefits obtained, so the divide is displaced rather than eliminated.

FIGURE 1 - THREE LEVELS OF DIVIDE

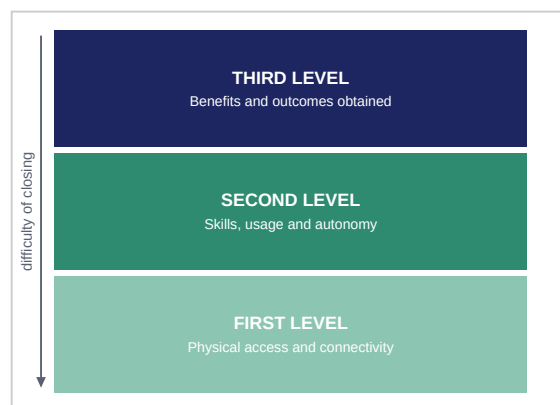


Figure 1. Closing a lower level displaces the inequality upward rather than removing it. This is why access-based measures show convergence while outcome-based measures do not, and why AI capability can widen inequality under conditions of universal access (after van Dijk, 2005).

HOW IT IS USED

Assess all three levels rather than access alone when appraising an AI-enabled service; universal access with unequal skills produces unequal outcomes and can widen rather than narrow the underlying inequality.

WHEN IT IS USED

In public-sector AI service design, in impact assessment of digital-only channels, in policy analysis of AI diffusion, and when a service is moved from a human to an automated channel.

BY WHOM IT IS USED

Public-sector service designers, policy analysts, corporate social responsibility and inclusion functions, and executives deciding to withdraw non-digital channels.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor framing was binary - connected or not - which made the problem appear soluble by infrastructure investment alone. The three-level model reconceives the divide as a sequence of displacements, each requiring a different intervention, and locates the durable inequality in outcomes rather than access. Applied to AI this predicts that a general-access technology can widen inequality if the capability to use it well is unequally distributed.

RELATED DEFINITIONS IN THIS COURSE

AI Literacy - Unit 2

Skill-Biased Technological Change - Unit 5

Externality - Unit 5

Algorithmic Bias - Unit 6

General Purpose Technology - Unit 5

SOURCES (HARVARD REFERENCING STYLE)

DiMaggio, P. and Hargittai, E. (2001) From the 'Digital Divide' to 'Digital Inequality'. Working Paper 15. Princeton, NJ: Center for Arts and Cultural Policy Studies.

Scheerder, A., van Deursen, A. and van Dijk, J. (2017) 'Determinants of internet skills, uses and outcomes: a systematic review of the second- and third-level digital divide', *Telematics and Informatics*, 34(8), pp. 1607-1624.

van Dijk, J.A.G.M. (2005) *The Deepening Divide: Inequality in the Information Society*. Thousand Oaks, CA: Sage.

AI Ethics

GLOSSARY 06 - SHEET 1 / 8

The applied field concerned with the moral acceptability of designing and deploying AI

PROFESSIONAL DEFINITION

AI ethics is the field of applied ethics concerned with the moral acceptability of the design, development, deployment and use of artificial intelligence systems, and with the obligations of those who commission and operate them. In the European context it is operationalised through the four principles of the Ethics Guidelines for Trustworthy AI - respect for human autonomy, prevention of harm, fairness and explicability - and their seven implementing requirements (HLEG AI, 2019; Floridi and Cowls, 2019).

WHO DEFINED IT ORIGINALLY

The four-principle formulation was adopted by the European Commission's High-Level Expert Group on Artificial Intelligence in 2019. Luciano Floridi of the University of Oxford and Josh Cowls (2019) derived a closely related unified set of five principles from a comparative analysis of six major ethics declarations, adding explicability to the four principles of biomedical ethics established by Beauchamp and Childress (1979).

WHY THIS DEFINITION WAS DEVELOPED

By 2018 dozens of organisations had published AI ethics principles, with substantial overlap and no common vocabulary, which made compliance claims incomparable. Floridi and Cowls demonstrated the convergence and supplied a single set; the High-Level Expert Group then attached implementing requirements, converting principles into an assessable form.

FIGURE 1 - FOUR PRINCIPLES, SEVEN REQUIREMENTS

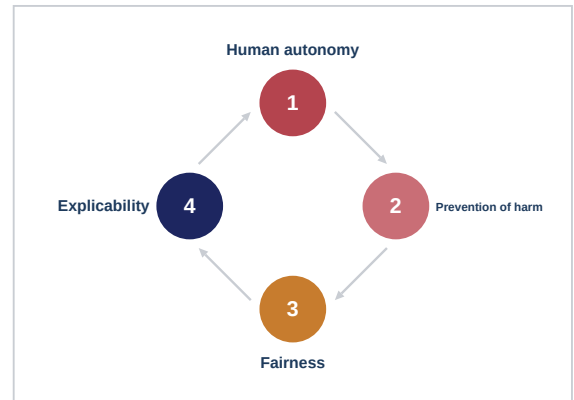


Figure 1. The four principles are not independently satisfiable. Explicability requirements can conflict with harm prevention where disclosure creates risk, and the auditable artefact is the recorded reasoning about how the conflict was resolved (after HLEG AI, 2019; Floridi and Cowls, 2019).

HOW IT IS USED

Apply the seven requirements as an assessment instrument during design rather than as a review at release, and record the trade-offs made between principles, since the principles conflict in practice and the record of how conflict was resolved is the auditable artefact.

WHEN IT IS USED

At the design stage of any consequential AI system, at ethics review board submission, in vendor assessment, and where two principles cannot be simultaneously satisfied.

BY WHOM IT IS USED

AI ethics boards and review committees, product owners and designers, risk and compliance functions, and executives accountable for deployment decisions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor instruments were general corporate codes of conduct, which were addressed to employee behaviour and gave designers no guidance about system properties. The trustworthy AI framework specifies requirements at the level of the system - human agency, technical robustness, privacy, transparency, fairness, societal wellbeing, accountability - and pairs them with an assessment list. Its acknowledged limitation is that it remains voluntary; the AI Act converts a subset into binding obligation, but the ethical requirements exceed the legal ones.

RELATED DEFINITIONS IN THIS COURSE

[Trustworthy AI - Unit 6](#)
[EU AI Act - Unit 6](#)
[Algorithmic Bias - Unit 6](#)
[Accountability Gap - Unit 6](#)
[Externality - Unit 5](#)

SOURCES (HARVARD REFERENCING STYLE)

Beauchamp, T.L. and Childress, J.F. (1979) *Principles of Biomedical Ethics*. New York: Oxford University Press.
 Floridi, L. and Cowls, J. (2019) 'A unified framework of five principles for AI in society', *Harvard Data Science Review*, 1(1).
 High-Level Expert Group on Artificial Intelligence (2019) *Ethics Guidelines for Trustworthy AI*. Brussels: European Commission.
 Jobin, A., Ienca, M. and Vayena, E. (2019) 'The global landscape of AI ethics guidelines', *Nature Machine Intelligence*, 1(9), pp. 389-399.

EU AI Act

The European Union's horizontal, risk-tiered regulation of artificial intelligence systems

GLOSSARY 06 - SHEET 2 / 8

PROFESSIONAL DEFINITION

The EU AI Act is Regulation (EU) 2024/1689, a horizontal regulation applying to any AI system placed on the Union market or whose output is used in the Union, regardless of the provider's place of establishment. It assigns obligations according to four risk tiers - unacceptable, high, limited and minimal - with separate obligations for general-purpose AI models, and it distinguishes the duties of providers from those of deployers (European Parliament and Council, 2024).

WHO DEFINED IT ORIGINALLY

Proposed by the European Commission in April 2021, negotiated between the European Parliament and the Council, and adopted as Regulation (EU) 2024/1689, published in the Official Journal on 12 July 2024. Its risk-based architecture draws on the New Legislative Framework for product safety and, in the academic commentary, on Veale and Zuiderveen Borgesius (2021).

WHY THIS DEFINITION WAS DEVELOPED

The Union sought an instrument that would neither suppress low-risk innovation nor leave dangerous applications unregulated, and that would apply across sectors rather than requiring separate rules for medical devices, finance and employment. Tiering the obligations by risk concentrated the compliance burden where harm potential is greatest.

FIGURE 1 - THE FOUR-TIER RISK PYRAMID



Figure 1. Obligations scale with the tier, and the great majority of AI systems fall in the lowest tier. Classification is by intended purpose and context of use, not by the technology employed, so the same model may sit in different tiers in different applications (after Regulation (EU) 2024/1689).

HOW IT IS USED

Classify each system by tier and by role - provider or deployer - then apply the corresponding duties: prohibition, conformity assessment and post-market monitoring, transparency obligations, or none.

WHEN IT IS USED

At AI project intake, before procurement, at any material modification or repurposing of a system, and on the phased application dates as successive provisions enter into force.

BY WHOM IT IS USED

Legal and compliance functions, chief AI officers and risk managers, product owners of AI systems, procurement teams assessing vendor conformity, and national market surveillance authorities.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor approach combined sector-specific regulation with non-binding principles such as the OECD Recommendation. The Act is horizontal, so a single classification serves across sectors, and it is binding and enforceable, with penalties for prohibited practices reaching the higher of a fixed ceiling or a percentage of worldwide annual turnover. Its extraterritorial reach, following the GDPR model, makes it consequential for non-EU providers whose output is used in the Union.

RELATED DEFINITIONS IN THIS COURSE

Artificial Intelligence - Unit 1

AI Governance - Unit 6

Trustworthy AI - Unit 6

AI Literacy - Unit 2

Human-in-the-Loop - Unit 2

SOURCES (HARVARD REFERENCING STYLE)

European Commission (2021) Proposal for a Regulation laying down harmonised rules on artificial intelligence, COM(2021) 206 final. Brussels: European Commission.

European Parliament and Council (2024) Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence. Official Journal of the European Union, L, 12 July 2024.

Veale, M. and Zuiderveen Borgesius, F. (2021) 'Demystifying the draft EU Artificial Intelligence Act', Computer Law Review International, 22(4), pp. 97-112.

Verification note: this term also appears as a framework sheet in MARTI301 Slibrary 1. The Act applies in phases, with different provisions entering into force on different dates, and implementing acts and harmonised standards continue to be issued. Verify the applicable dates and the current text of any Article against EUR-Lex before relying on it in assessed work.

Algorithmic Bias

Systematic and unfair discrimination in the outputs of an algorithmic system

GLOSSARY 06 - SHEET 3 / 8

PROFESSIONAL DEFINITION

Algorithmic bias is systematic and repeatable error in the output of a computational system that creates unfair outcomes, such as privileging one group of users over another. It arises from pre-existing bias embedded in the training data or in social practice, technical bias arising from the design of the system, and emergent bias arising when a system is used in a context different from the one for which it was built (Friedman and Nissenbaum, 1996; Mehrabi et al., 2021).

WHO DEFINED IT ORIGINALLY

Batya Friedman and Helen Nissenbaum established the three-part taxonomy in ACM Transactions on Information Systems in 1996. The contemporary empirical literature is anchored by Joy Buolamwini and Timnit Gebru's (2018) Gender Shades audit of commercial facial-analysis systems; Ninareh Mehrabi and colleagues (2021) provide the systematic survey of sources and mitigations.

WHY THIS DEFINITION WAS DEVELOPED

Automated systems were presented as neutral because they were mechanical, which obscured the fact that they encode the distribution of the data used to build them and the assumptions of those who built them. Friedman and Nissenbaum needed a taxonomy that distinguished sources, because each source requires a different remedy and the same observed disparity may have any of the three.

HOW IT IS USED

Audit outputs by subgroup rather than in aggregate, since aggregate accuracy can be high while subgroup accuracy is not; then identify which of the three sources is operating, because data rebalancing does not fix a technical or emergent bias.

WHEN IT IS USED

Before deployment of any system affecting individuals, at periodic audit of a live system, when a subgroup disparity is reported, and whenever a system is used in a new population or context.

BY WHOM IT IS USED

Model validation and audit functions, data scientists selecting training data, legal and compliance teams assessing discrimination exposure, and deployers of vendor systems built on unknown data.

FIGURE 1 - THREE SOURCES OF BIAS

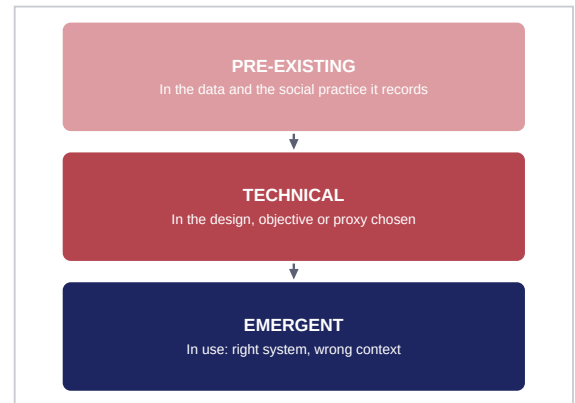


Figure 1. The three sources require different remedies and can produce identical observed disparities. Rebalancing the training data addresses only the first, which is why audits that stop at data composition frequently fail to remove the disparity (after Friedman and Nissenbaum, 1996).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor treatment framed bias as a statistical property of an estimator, which addressed variance from a true value but not fairness across groups. The Friedman and Nissenbaum taxonomy relocates the question to sources and to social consequence, and identifies emergent bias - correct system, wrong context - which purely statistical framings cannot express. Buolamwini and Gebru's disaggregated audit method then converted the argument into a measurement protocol that could be run on a commercial system from the outside.

RELATED DEFINITIONS IN THIS COURSE

[AI Ethics - Unit 6](#)
[Explainability - Unit 6](#)
[AI Governance - Unit 6](#)
[Digital Divide - Unit 5](#)
[Trustworthy AI - Unit 6](#)

SOURCES (HARVARD REFERENCING STYLE)

- Buolamwini, J. and Gebru, T. (2018) 'Gender shades: intersectional accuracy disparities in commercial gender classification', Proceedings of Machine Learning Research, 81, pp. 77-91.
- Friedman, B. and Nissenbaum, H. (1996) 'Bias in computer systems', ACM Transactions on Information Systems, 14(3), pp. 330-347.
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K. and Galstyan, A. (2021) 'A survey on bias and fairness in machine learning', ACM Computing Surveys, 54(6), pp. 1-35.

Explainability

The property of a system whose outputs can be accounted for in humanly intelligible terms

GLOSSARY 06 - SHEET 4 / 8

PROFESSIONAL DEFINITION

Explainability is the property of an AI system whose behaviour and specific outputs can be accounted for in terms intelligible to a relevant human audience. It is distinguished from interpretability, which concerns whether the model's internal mechanism is directly inspectable, and it is audience-relative: the explanation adequate for a data scientist differs from that required by a decision subject or a regulator (Guidotti et al., 2018; Rudin, 2019).

WHO DEFINED IT ORIGINALLY

The research field was named and funded through DARPA's Explainable AI programme from 2016. Riccardo Guidotti and colleagues (2018) supplied the systematic taxonomy of methods in ACM Computing Surveys. Cynthia Rudin of Duke University (2019) argued in Nature Machine Intelligence that for high-stakes decisions inherently interpretable models should be used instead of post hoc explanations of opaque ones.

WHY THIS DEFINITION WAS DEVELOPED

Accurate models were being deployed in consequential decisions without any account of the basis for individual outcomes, which prevented contest, correction and audit. Explainability was needed to make the individual decision reviewable; Rudin's intervention added the argument that a post hoc explanation of an opaque model is an approximation of the model rather than the reason for the decision.

HOW IT IS USED

Specify the audience and the purpose before selecting a method, and prefer an inherently interpretable model where the decision is consequential, since a post hoc explanation may be faithful to the model's behaviour without being the actual reason for the outcome.

WHEN IT IS USED

At the design of any system whose outputs affect individuals, when a decision must be contested or appealed, in conformity assessment of a high-risk system, and in model validation.

BY WHOM IT IS USED

Data scientists selecting model families, model risk and validation functions, compliance teams evidencing transparency duties, and the decision subjects entitled to an account.

FIGURE 1 - EXPLANATION BY AUDIENCE AND SCOPE

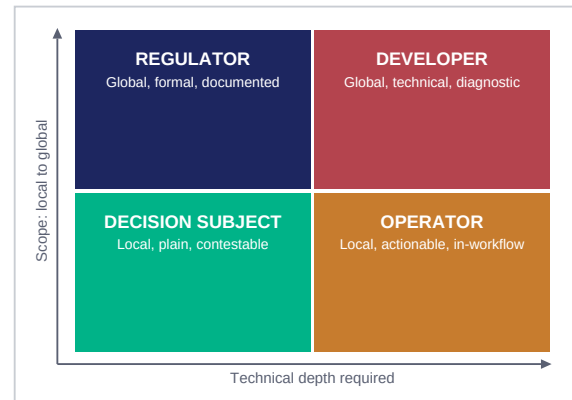


Figure 1. Explainability is audience-relative and no single artefact serves all four. A feature-importance plot satisfies the developer and is not an account of why this applicant was refused, which is what the decision subject is entitled to (after Guidotti et al., 2018).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor position accepted opacity as the price of accuracy and offered global feature-importance measures as a substitute for individual explanation. The field's contribution is to distinguish global from local explanation and model-agnostic from model-specific methods, so the right instrument can be matched to the question asked. Rudin's argument then challenges the accuracy-interpretability trade-off itself, holding that on structured data the trade-off is frequently negligible and the opacity therefore unnecessary.

RELATED DEFINITIONS IN THIS COURSE

Algorithmic Bias - Unit 6

Accountability Gap - Unit 6

Human-in-the-Loop - Unit 2

AI Governance - Unit 6

Algorithm Aversion - Unit 2

SOURCES (HARVARD REFERENCING STYLE)

- Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F. and Pedreschi, D. (2018) 'A survey of methods for explaining black box models', ACM Computing Surveys, 51(5), pp. 1-42.
- Gunning, D. and Aha, D.W. (2019) 'DARPA's explainable artificial intelligence program', AI Magazine, 40(2), pp. 44-58.
- Rudin, C. (2019) 'Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead', Nature Machine Intelligence, 1(5), pp. 206-215.

AI Governance

GLOSSARY 06 - SHEET 5 / 8

The structures and processes by which an organisation directs and controls its AI activity

PROFESSIONAL DEFINITION

AI governance is the system of structures, decision rights, policies, processes and controls by which an organisation directs its AI activity and holds itself accountable for the outcomes. In the standardised form it comprises four functions - govern, map, measure and manage - of which govern is a culture and accountability function running across the other three rather than a preliminary stage (NIST, 2023; ISO/IEC, 2023).

WHO DEFINED IT ORIGINALLY

The AI Risk Management Framework was published by the United States National Institute of Standards and Technology in January 2023 following a public consultation process. ISO/IEC 42001:2023, the AI management system standard, provides the certifiable counterpart. Both were designed to align with the emerging EU regulatory requirements without being specific to them.

WHY THIS DEFINITION WAS DEVELOPED

Organisations were managing AI risk within existing model-risk or IT-governance functions, which addressed neither the data dependency nor the sociotechnical harms that AI systems generate. A dedicated framework was needed that was voluntary, sector-neutral and capable of being mapped onto multiple regulatory regimes, since firms operate across jurisdictions with divergent rules.

HOW IT IS USED

Map the AI system inventory and its context, measure the risks with defined metrics, manage them through prioritised controls, and treat govern as the continuous accountability layer in which decision rights and escalation paths are fixed.

WHEN IT IS USED

At the establishment of an AI function, before high-risk deployment, at periodic risk review, in preparation for conformity assessment or certification, and after any incident.

BY WHOM IT IS USED

Boards and audit committees, chief risk officers and AI governance leads, internal audit testing the controls, and business owners of individual systems.

FIGURE 1 - FOUR GOVERNANCE FUNCTIONS

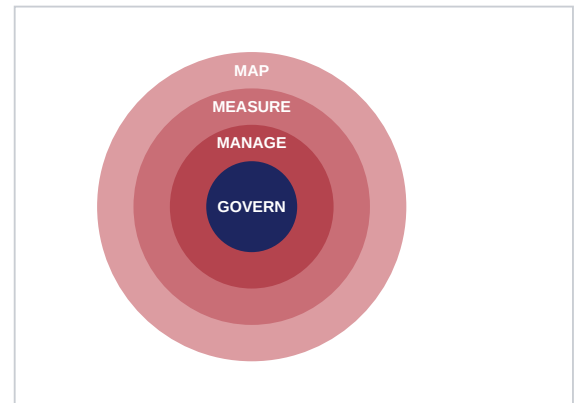


Figure 1. Govern is drawn at the centre because it is a continuous accountability function rather than a first step. Where decision rights and escalation paths are undefined, mapping, measurement and management produce documentation without control (after NIST, 2023).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Its predecessor was model risk management as codified for banking - notably the Federal Reserve's SR 11-7 - which assumed a documented, statistically validated model with an identifiable owner and a stable input distribution. The AI framework extends governance to data provenance, sociotechnical harm, third-party and general-purpose models, and continuous post-deployment monitoring, none of which the model-risk tradition addressed. Its structure as a voluntary framework mappable to several regimes is what allows one control set to serve multiple jurisdictions.

RELATED DEFINITIONS IN THIS COURSE

[EU AI Act - Unit 6](#)
[Decision Rights - Unit 2](#)
[MLOps - Unit 4](#)
[Accountability Gap - Unit 6](#)
[AI Operating Model - Unit 3](#)

SOURCES (HARVARD REFERENCING STYLE)

International Organization for Standardization and International Electrotechnical Commission (2023) ISO/IEC 42001:2023 Information Technology - Artificial Intelligence - Management System. Geneva: ISO.

National Institute of Standards and Technology (2023) Artificial Intelligence Risk Management Framework (AI RMF 1.0), NIST AI 100-1. Gaithersburg, MD: NIST.

Board of Governors of the Federal Reserve System (2011) Supervisory Guidance on Model Risk Management, SR 11-7. Washington, DC: Federal Reserve.

Accountability Gap

GLOSSARY 06 - SHEET 6 / 8

The condition in which harm occurs and no party can properly be held responsible

PROFESSIONAL DEFINITION

The accountability gap - also termed the responsibility gap - is the condition in which an autonomous or learning system causes harm and no human party satisfies the conditions ordinarily required for moral or legal responsibility, because no one had the control or the foreseeability that responsibility presupposes. Many-hands problems in distributed development compound it, as responsibility diffuses across designers, deployers, operators and data providers (Matthias, 2004; Santoni de Sio and Mecacci, 2021).

WHO DEFINED IT ORIGINALLY

Andreas Matthias identified and named the problem in Ethics and Information Technology in 2004. Filippo Santoni de Sio and Giulio Mecacci (2021) decomposed it into four distinct gaps - culpability, moral accountability, public accountability and active responsibility - in Philosophy and Technology. Helen Nissenbaum's (1996) work on many hands in computing is the direct antecedent.

WHY THIS DEFINITION WAS DEVELOPED

Traditional responsibility ascription requires that an agent could have known and could have done otherwise. Systems that learn after deployment weaken both conditions, so harms could occur for which no one was answerable. Naming the gap made it a design and governance problem - to be closed by allocation - rather than an unavoidable feature of the technology.

HOW IT IS USED

Assign a named accountable person for each deployed system before deployment, ensure that person has both the information and the authority to act, and record the assignment; a gap identified after harm has occurred cannot be closed retrospectively.

WHEN IT IS USED

Before deployment of any autonomous or consequential system, at incident review, in vendor contracting where the model is supplied by a third party, and in insurance and liability assessment.

BY WHOM IT IS USED

Boards accepting residual risk, general counsel and risk officers, system owners named as accountable, and regulators determining where liability rests.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor position was product liability, which attaches to a defect in an artefact with fixed properties at the point of sale. A system that adapts after deployment may cause harm without any defect at the point of supply, which product liability was not constructed to address. Santoni de Sio and Mecacci's four-way decomposition supplies the practical advance: the four gaps close by different means - contractual allocation, organisational design, transparency duties and meaningful human control - so the remedy depends on which gap is actually open.

RELATED DEFINITIONS IN THIS COURSE

AI Governance - Unit 6

Human-in-the-Loop - Unit 2

Explainability - Unit 6

Decision Rights - Unit 2

EU AI Act - Unit 6

SOURCES (HARVARD REFERENCING STYLE)

- Matthias, A. (2004) 'The responsibility gap: ascribing responsibility for the actions of learning automata', Ethics and Information Technology, 6(3), pp. 175-183.
- Nissenbaum, H. (1996) 'Accountability in a computerized society', Science and Engineering Ethics, 2(1), pp. 25-42.
- Santoni de Sio, F. and Mecacci, G. (2021) 'Four responsibility gaps with artificial intelligence: why they matter and how to address them', Philosophy and Technology, 34(4), pp. 1057-1084.

FIGURE 1 - FOUR GAPS, FOUR REMEDIES



Figure 1. The four gaps are distinct and close by different means. Naming an accountable owner addresses the second and does nothing for the fourth, which requires that the named person hold both the information and the authority to intervene (after Santoni de Sio and Mecacci, 2021).

Data Protection by Design

Embedding data protection into processing from the outset and by default

GLOSSARY 06 - SHEET 7 / 8

PROFESSIONAL DEFINITION

Data protection by design and by default is the legal obligation, under Article 25 of the General Data Protection Regulation, to implement appropriate technical and organisational measures that give effect to the data-protection principles at the time processing means are determined and during processing itself, and to ensure that by default only personal data necessary for each specific purpose is processed. It converts a design philosophy into a binding duty on the controller (Regulation (EU) 2016/679, Article 25; Cavoukian, 2009).

WHO DEFINED IT ORIGINALLY

The privacy by design philosophy and its seven foundational principles were formulated by Ann Cavoukian as Information and Privacy Commissioner of Ontario in the 1990s and set out formally in 2009, receiving international recognition at the 2010 conference of data protection commissioners. It was made a legal obligation by Article 25 of the GDPR, applicable from 25 May 2018.

WHY THIS DEFINITION WAS DEVELOPED

Privacy protection was applied at the end of development as a compliance review, by which point the architecture had already determined what data would be collected and retained, and remediation was expensive or impossible. Moving the obligation to the point at which processing means are determined makes protection an architectural constraint rather than a subsequent review.

HOW IT IS USED

Determine the lawful basis, minimise the data and set the retention period before the architecture is fixed; configure the default to the most protective setting, since Article 25 requires protection by default rather than as a user-selected option.

WHEN IT IS USED

At the determination of processing means, before an AI system is trained on personal data, at data protection impact assessment, and at any change of processing purpose.

BY WHOM IT IS USED

Data protection officers, system architects and engineers, product owners defining processing purposes, and controllers who bear the legal obligation.

FIGURE 1 - WHERE PROTECTION IS DETERMINED

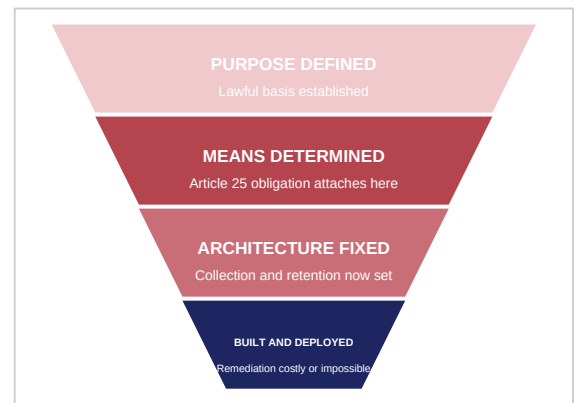


Figure 1. The obligation attaches at the second stage, before the architecture constrains what is possible. A data-protection review conducted at the fourth stage can document the position but can no longer change it (after Regulation (EU) 2016/679, Article 25).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor practice was notice and consent, which placed the burden on the individual to read a policy and accept it, and which permitted extensive collection provided it was disclosed. Article 25 shifts the burden to the controller and makes minimisation a default state rather than an option, so the protective configuration obtains without any action by the data subject. For AI this bites at the point of training-data assembly, where the temptation to retain everything is strongest.

RELATED DEFINITIONS IN THIS COURSE

Data Strategy - Unit 3

AI Governance - Unit 6

EU AI Act - Unit 6

Trustworthy AI - Unit 6

Algorithmic Bias - Unit 6

SOURCES (HARVARD REFERENCING STYLE)

Cavoukian, A. (2009) Privacy by Design: The 7 Foundational Principles. Toronto: Information and Privacy Commissioner of Ontario.
European Data Protection Board (2020) Guidelines 4/2019 on Article 25 Data Protection by Design and by Default, version 2.0. Brussels: EDPB.
European Parliament and Council (2016) Regulation (EU) 2016/679 (General Data Protection Regulation), Article 25. Official Journal of the European Union, L119, 4 May 2016.

Trustworthy AI

AI that is lawful, ethical and technically robust across its whole lifecycle

GLOSSARY 06 - SHEET 8 / 8

PROFESSIONAL DEFINITION

Trustworthy AI is artificial intelligence that satisfies three components throughout its lifecycle: it is lawful, complying with applicable law; ethical, respecting ethical principles and values; and robust, both technically and socially, since even well-intentioned systems can cause unintentional harm. The three components are necessary and not individually sufficient, and the framework is operationalised through seven requirements with an accompanying assessment list (HLEG AI, 2019).

WHO DEFINED IT ORIGINALLY

The framework was set out by the European Commission's High-Level Expert Group on Artificial Intelligence in the Ethics Guidelines for Trustworthy AI (April 2019), with the Assessment List for Trustworthy AI following in July 2020. It provided the conceptual foundation for the White Paper on AI (2020) and, subsequently, for the AI Act.

WHY THIS DEFINITION WAS DEVELOPED

Ethics guidance was proliferating without any bridge to law or to engineering practice, so ethical commitments and technical assurance ran on separate tracks. Combining lawfulness, ethics and robustness in a single definition made the point that legal compliance alone is insufficient and that ethical intent without technical robustness produces harm regardless.

FIGURE 1 - THREE NECESSARY COMPONENTS

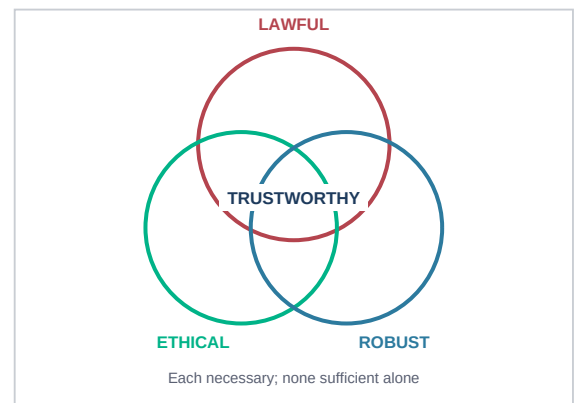


Figure 1. A system may be lawful and ethical and still cause harm through technical fragility; it may be robust and lawful and still be unacceptable. Only the intersection satisfies the definition, and the framework requires that conflicts between the three be recorded rather than resolved silently (after HLEG AI, 2019).

HOW IT IS USED

Assess the system against the seven requirements using the assessment list at design, at deployment and at review; record the trade-offs made between requirements and treat the record as the evidence of due diligence.

WHEN IT IS USED

At AI system design, before market placement, at periodic review of a live system, in vendor assessment, and in preparation for conformity assessment under the AI Act.

BY WHOM IT IS USED

AI ethics and governance boards, product and engineering leads, procurement functions assessing vendors, and executives accountable for the deployment decision.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor instruments were principle declarations, which stated commitments without any operational test, and technical assurance standards, which tested robustness without reference to ethical or legal acceptability. The trustworthy AI framework joins the three and supplies an assessment list, converting a statement of values into an auditable process. Its acknowledged weakness is that it does not resolve conflicts between the requirements; it requires only that trade-offs be made explicitly and recorded.

RELATED DEFINITIONS IN THIS COURSE

AI Ethics - Unit 6

EU AI Act - Unit 6

AI Governance - Unit 6

Explainability - Unit 6

Human-in-the-Loop - Unit 2

SOURCES (HARVARD REFERENCING STYLE)

High-Level Expert Group on Artificial Intelligence (2019) Ethics Guidelines for Trustworthy AI. Brussels: European Commission.

High-Level Expert Group on Artificial Intelligence (2020) The Assessment List for Trustworthy Artificial Intelligence (ALTAI). Brussels: European Commission.

European Commission (2020) White Paper on Artificial Intelligence: A European Approach to Excellence and Trust, COM(2020) 65 final. Brussels: European Commission.